

SMOOTHING INTERMEDIATE QUANTILES

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Abstract

We expand the toolbox for analyzing extreme values by extending the notion of smoothed quantile estimation to quantile levels converging to the right-tail. The classical tools for analyzing high-values rely on performing quantile estimation, which involves optimizing a piecewise linear objective function. Recent work by Fernandes, Guerre, and Horta (2021) ameliorated this problem by using convolution smoothing. We expand on their work by expanding the class of kernels under consideration and allowing the quantile level to diverge. We establish consistency and asymptotic normality of our estimator under appropriate assumptions. We verify the validity of our estimator through a Monte Carlo study and apply it to estimate low quantiles of birthweights.

Contents

1	Introduction	3
2	Literature Overview	7
2.1	Quantile Estimation	8
2.2	Smoothing quantile estimation	9
2.3	Extreme value Theory	11

3	Methodology	14
4	Simulations	19
4.1	Theoretical Verification	20
4.2	Breaking Bandwidth	25
4.3	Asymptotic bias	29
5	Application	35
6	Conclusion	38
A	Proofs	43
B	Generative AI statement	48

Popular Summmary

Large values in data are often hard to interpret and make sense of. There are different answers to this problem. We can consider such values to be "outliers" and seek to find ways to minimize their influence on the final results. This is known as the field of robust statistics. In contrast, we consider these values to be part of the distribution, and we aim to provide inference on precisely these values. This is particularly relevant for fields where large values occur, such as finance, insurance, meteorology, earthquakes, internet congestion, and more. The field of extreme value theory is centered on these issues. However, we require a definition, in general, what "high" means. For example, for a uniform distribution on $[0, 1]$, being "high" could mean being very close to 1, while this would still be very central for a normal distribution. We utilize the notion of quantiles for this, which corresponds to the smallest value such that a given probability level is attained. The problem is that, in general, estimating a quantile requires minimizing a non-smooth objective function. This precludes the use of first and second order gradients and Newtonian methods, which can make life a lot easier. Recent work by Fernandes, Guerre, and Horta (2021) has found a way to restore the smoothness of the objective function while retaining the quantile estimate. However, their work restricts the type of smoothness they can consider and is unable to model large values, corresponding to high quantiles. We expand on their work by broadening the class of kernels that can be utilized and the range that the quantile level can take. We show that the estimators concentrate around the true value and provide distributional limit results to allow for inference using classical tools. Further, we show that the classical kernel density estimator can be utilized as an unbiased estimator for the theoretical scaling of the estimator, a new result within extreme value theory.

1 Introduction

There has been a decrease in child infant mortality in the United States (US) over the last century (Park et al., 2025). This trend has recently seen a break (Gruppuso, Oster, and Adashi, 2025), where infant mortality rose by 3% in 2022. There are a number of possible factors that could explain this phenomena, with the prime candidate being the covid 19 pandemic. However, the report of Gruppuso, Oster, and Adashi (2025) cautions against this conclusion, as there is no clear data to support this. Alternative explanations could be that social-economic unrest exacerbated existing inequalities, making existing differences in child mortality worse (Singh and Stella, 2019). To investigate this trend, we note that there exists a clear relation between infant birth weight and infant mortality (Jana et al., 2023).

The inference on birthweights can be enhanced by looking beyond means and averages. The full distribution will provide a clearer answer to where this trend break came from. Figure 1 shows the empirical distribution of birthweights, where we plot the empirical probability against the order statistics. We see that the median is achieved at approximately 3500 grams, with a relatively steep increase in the probability, denoting more weight in the middle.

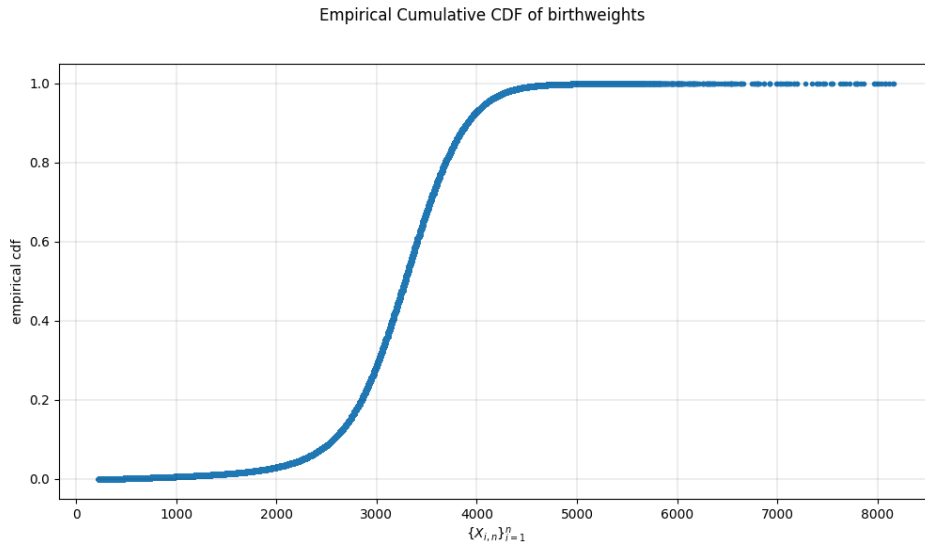


Figure 1: Plot of the order statistics of the birthweight against their approximate probability distribution.

However, this figure does not tell anything the birthweights we are interested in, the low ones. We want to be able to take the lowest values and provide inference on those. To this end, consider Figure 2, which displays the lowest order statistics. These plots indicate that the minimum is around 215 grams. However, more importantly, it shows that the distribution converges towards the left-end point in approximately the same manner when we consider lower and lower order statistics. This suggests that there is a certain self-similarity within the tail. In particular, it suggests that the same behavior governs the data as you investigate the lowest

observed values. If this were true, this would mean that we can extrapolate outward from intermediate quantiles into the tails.

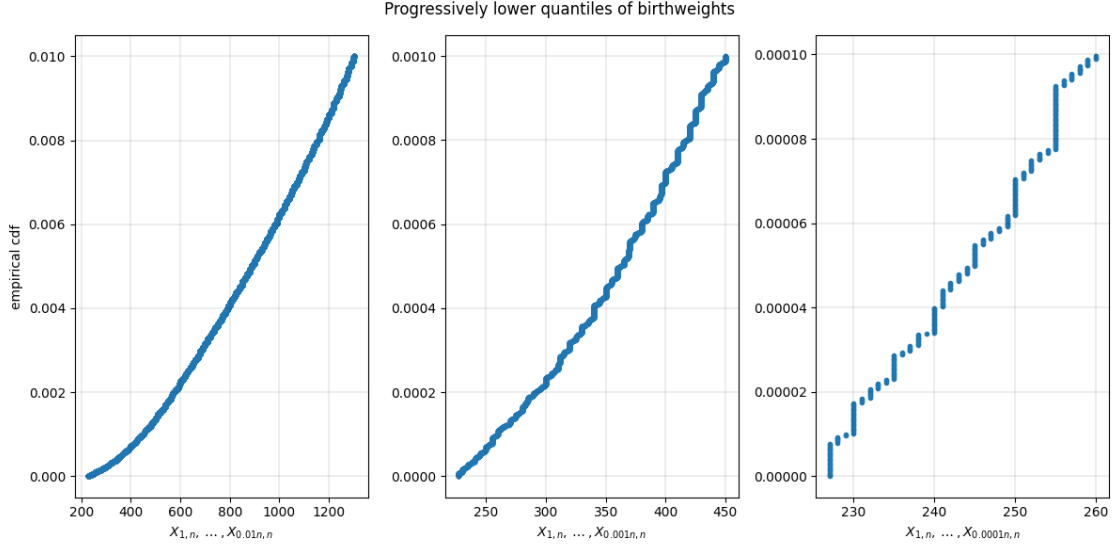


Figure 2: Lowest quantiles of the birthweights of babies in 2024.

The lowest birthweights suffer from a sparsity of data. This is by nature of the problem, as any distribution only rarely draws data from its tails, even for distributions with a finite right- or left-endpoint. We want to be able to make valid inferences on this model. Hence, we formalize the self-similarity assumption mentioned before, which is known as *regular variation* within the extreme value community.

The monograph of Haan and Ferreira (2006) provides a clear overview of this model assumption and the literature that grew from it. Similarly to the central limit theorem, the goal is to formulate a "max" (or "min") limit theorem. For a sample $\{Z_i\}_{i=1}^n$ distributed according to the random variable Z with cdf F_Z , we assume that there exists $b_n \in \mathbb{R}$ and $a_n > 0$ for $n \in \mathbb{N}$ such that

$$\frac{\max\{X_1, \dots, X_n\} - b_n}{a_n}, \quad (1)$$

has a non-degenerate limit. This is equivalent to showing that

$$\mathbb{P}\left(\frac{\max\{X_1, \dots, X_n\} - b_n}{a_n} \leq x\right) = F^n(a_n x + b_n) \rightarrow G(x), \quad n \rightarrow \infty, \quad (2)$$

for some distribution G . By the famed Fisher-Tippet-Gnedenko Theorem (see Theorem 1.1.3 of (Haan and Ferreira, 2006)), we know that this limit has an analytic form given by a single parameter, which is known as the first order extreme value parameter.

The assumption given in (2) ensures that the maxima and/or minima do not have a degenerate distribution in the limit after appropriate normalization. This

assumption ensures that the minima and low order statistics of our data provide a closed form distribution as we consider lower and lower order statistics. Furthermore, as the extreme value condition provides structure on the entire tail, we can perform inference on the observed maxima and beyond. However, to perform inference on the data, we need a "jumping-off" point which is known as the intermediate quantile in the extreme value literature. Hence, we need to be able to perform inference on this quantile before anything else. To this end, we define the estimated quantile as the solution to the minimization problem

$$\hat{R}(\psi; \tau) = \frac{1}{n} \sum_{i=1}^n \rho_{\tau}(Z_i - \psi), \quad \tau \in [0, 1],$$

for a sample $\{Z_i\}_{i=1}^n$ with the pinball loss-function $\rho_{\tau}(u) = u(\tau - \mathbb{1}\{u < 0\})$. It is a classic result for $\tau \in (0, 1)$ that the solution to this minimization problem is a consistent and asymptotically normal estimator for the true quantile $F^{\leftarrow}(\tau)$, see Van der Vaart (2000).

However, it is important to note that this result only holds for $\tau \in (0, 1)$ or a compact subset of $(0, 1)$. This precludes the very low order statistics and the minima we are interested in. Hence, we differentiate between the different levels of quantiles, as in Chernozhukov, Fernández-Val, and Kaji (2017).

They define (a) *central* quantiles as those for which their level $\tau \in (0, 1)$, i.e. bounded away from the $\{0, 1\}$ boundary. They define (b) *intermediate* quantiles as those for which their level satisfies $\tau_n \rightarrow 0$ as $n \rightarrow \infty$ and $n\tau_n \rightarrow \infty$ as $n \rightarrow \infty$. This is often defined through the number of exceedances, $k = n\tau_n \in \mathbb{N}$. Finally, they define (c) *extreme* quantiles as those for which $\tau_n \rightarrow 0$ and $n\tau_n \rightarrow c > 0$ as $n \rightarrow \infty$. Additionally, they show that the inference for these types of non-central quantiles differ from the traditional central quantiles. The rate of convergence changes to depend on the τ_n variable, and the difference between the true and estimated quantiles need to be scaled properly to avoid divergence.

The classic quantile estimator has a piecewise linear loss function. This creates difficulties when working with the estimate, as the non-continuity of the risk function creates a non-continuous estimate w.r.t the quantile levels, and the optimization of the quantile risk function requires more sophisticated tools than off-the-shelf methods. This can lead to a slow-down of the estimator when compared to classic results.

An alternative has been proposed in Nadaraya (1964), which smooths the error of the quantile estimate in the risk function through a kernel function, i.e.

$$\hat{R}_h(\psi; \tau) = \frac{1}{n} \sum_{i=1}^n \int_{-\infty}^{\infty} \rho_{\tau}(u) K_h(u - (Z_i - \psi)) du, \quad \tau \in [0, 1],$$

where $K_h(x) = \frac{1}{h} K\left(\frac{x}{h}\right)$ is the rescaled kernel function, see Condition 1. The work of Nadaraya (1964) established consistency and asymptotic normality, where the more recent work of Fernandes, Guerre, and Horta (2021) generalized this to quantile regression. These results work well on the compact set of $(0, 1)$, but they fall short for the intermediate and extreme quantiles, which we are interested in. Thus, the goal of this thesis is to generalize these results to the cases intermediate and extreme

quantile regression. We aim to provide a complete framework for inference on these quantiles, establishing consistency, asymptotic normality, and the consistency of the estimator for the variance. This last point is the main addition to the literature, providing a possible improvement over the self-normalized quantiles of Chernozhukov, Fernández-Val, and Kaji (2016).

The structure of the thesis is as follows. We provide a brief overview on the literature of quantile regression, smoothing quantile regression, and extreme value theory in Section 2. In section 3, we show the consistency and asymptotic normality of our extreme quantile estimator under appropriate assumptions. In addition, we verify and expand the results through a Monte Carlo in Section 4. Finally, we conclude in Section 6 and provide further avenues of research.

Throughout the thesis, we consider the sample of independent and identically distributed (iid) data $\{Z_i\}_{i=1}^n$ generated according to the random variables Z with cumulative distribution function F . We define the associated empirical cdf as $\mathbb{F}_n$. We denote the order statistics as $Z_{1,n} \leq Z_{2,n} \leq \dots \leq Z_{n,n}$. For a measurable function f , we define the generalized left-inverse as

$$f^{\leftarrow}(z) = \inf\{x \in \mathbb{R} : f(x) \geq z\}.$$

We denote $\text{supp}(f)$ as the support for a function f . For a measurable function f , we define the rescaled function $f_h(z) := \frac{1}{h}f(z/h)$ for $h > 0$ and $z \in \text{supp}(f)$. We define $B_\varepsilon(x)$ as the closed interval around $x \in \mathbb{R}$ of length 2ε , i.e. $B_\varepsilon(x) = \{y \in \mathbb{R} : |y - x| \leq \varepsilon\}$.

2 Literature Overview

The distribution of a random variable Z with cdf F_Z can be described by a number of different characterizations. This includes the moments, such as the mean and variance for the normal distribution. However, these moments might be inadequate to describe certain classes of distributions. Heavy-tailed distributions are one such class, which might not have even have moments to describe these variables. For example, the Pareto distribution with distribution function $\mathbb{P}(Z > x) = \frac{1}{x}$ for $x > 1$ has no finite first or higher order moment. This complicates matters, as functions such as the moment generating function become less- or non-informative. Alternatively, one could consider the characteristic function, i.e. $t \mapsto \mathbb{E}[e^{itX}]$, which does not suffer from these problems. However, this function can be relatively hard to estimate and does not immediately provide a clear interpretation for heavy tails. The notion of quantiles can be very useful in this matter. They are defined as

$$Q_z(\tau) = F_Z^{\leftarrow}(\tau), \quad \tau \in [0, 1].$$

This denotes the value at which a certain probability mass is accumulated

The notion of quantiles provides a few advantages as opposed to traditional moment assumptions. The quantiles provide more flexibility and their existence require less stringent assumptions. Inference based on quantiles is more powerful, as they can characterize the entire distribution. This is particularly relevant for applications such as income inequality, birth weights, traffic deaths, precipitation, temperature, or market returns where one is often less interested in the mean of these variables and more in the low or high values.

The notion of quantiles provides a tool for data exploration of univariate data. Most packages for data exploration, such as the `pandas.DataFrame.summary()` method in Python, report the maximum and minimum of a dataset by default, which correspond to the highest and lowest empirical quantiles. This distribution of different probability levels provide a clear idea on the distribution. The difference between a mean and median of a data set is often a clear initial diagnosis of the tails of the distribution. Furthermore, the notion of quantiles is heavily used in finance. In particular, one of the main measures of risk written in law, the Basel III accords (Petrović and Trifunović, 2024) specifically, is the so-called Value-at-Risk (VaR) (Holton, 2002). The VaR is equal to the $1 - \tau$ quantile. This measure has been utilized heavily throughout the finance literature. Extensions incorporating slightly higher moments, such as in the Expected Shortfall (ES) (Tasche, 2002), have been proposed to provide alternatives to VaR. This has later become part of the more general literature on coherent risk measures within the finance space (Delbaen, 2002).

However, outside of finance, the relevance of quantiles can also be seen. In particular, the notion of quantiles for conditional distribution has made a large impact on the ability to do inference on different data sets. The advantages mentioned previously of quantiles over moment conditions have led the field of quantile regression to develop rapidly over time. The theoretical foundations for quantile regression and its relative simplicity has allowed the field to flourish with a wide diversity of applications and extensions into different domains. From a method-

ological perspective, these extensions range from causal effects modeling (S. Sun, Moodie, and Nešlehová, 2021), to time series (Katsouris, 2023), to panel data models (Canay, 2011), to even functional data analysis (M. Li et al., 2022). The review article by Koenker (2017) provides an overview of the applications within these different fields.

From an empirical perspective, quantile regression has been applied to a number of different fields. This includes applications in biomedical data (Mazucheli et al., 2022), economics (Chernozhukov, Fernández-Val, and Kaji, 2016), asset pricing (Maiti, 2021), insurance claim modeling (Heras, Moreno, and Vilar-Zanón, 2018), and engineering (Xu et al., 2023). Within these fields, the quantiles of the variable in question are often defined by the relevant material, such as the Value at Risk (VaR) in finance. However, the applications are not limited to inference based models. In contrast, the more recent field of conformal inference utilizes quantiles and conditional quantiles to evaluate the efficacy of general black box models, providing uncertainty quantification based solely on predictions under an exchangeability assumption. This allows for the uncertainty quantification of predictions from models where classical theory remains intractable.

2.1 Quantile Estimation

The classical problem of quantile estimation can be considered on the population level. We consider a continuous distribution for ease of notation. To find the quantile at level p of this distribution, we want to find ψ such that $F_Z(\psi) = p$. This can be seen as the first order condition for the minimization of

$$\begin{aligned} R(\psi; \tau) &:= \mathbb{E}[(Z - \psi)\mathbb{1}\{Z - \psi < 0\}(\tau - 1)] + \mathbb{E}[(Z - \psi)\mathbb{1}\{Z - \psi > 0\}\tau] \\ &= \mathbb{E}[\rho_\tau(Z - \psi)] = \int \rho_\tau(z - \psi) dF_Z(z), \end{aligned}$$

for the pin-ball loss $\rho_\tau(u) = u(\tau - \mathbb{1}\{u < 0\})$. However, as we only observe the sample $\{Z_i\}_{i=1}^n$, we need to approximate the risk function by an estimated risk function. This can be done relatively straightforwardly by replacing the true cdf F_Z by the empirical cdf $\mathbb{F}_n(z) := \frac{1}{n} \sum_{i=1}^n \mathbb{1}\{Z_i \leq z\}$. The results on this classical estimator are very well established, as they reduce to the order statistics. Furthermore, there is even a fast implementation using linear programming to solve the minimization of the empirical risk function.

The classical quantile estimation is well established and has a clear advantage in robustness. The quantile estimator is robust to slight perturbations of the underlying sample. In particular, suppose that we perturb either the value below or above a given value ψ by a small value. Then, by the first order condition, as long as none of the perturbed values below or above ψ cross this level, it remains the same. This is a relatively strong result, and indicates that outliers have far less of an effect on the estimator than the usual moment estimators.

However, this strength can simultaneously be a weakness. The piecewise linear loss function means that you have a non-smooth objective function. This necessitates the use of a linear programming approach as opposed to more standard off-the-shelf optimization techniques. Furthermore, theoretical results require a different

approach to most optimization procedures with a smooth objective function, complicating matters from a technical standpoint. Finally, the non-smoothness of the risk function also means that the quantile estimate is non-smooth in the quantile levels. This means that constructing the entire quantile function $\tau \mapsto Q_Z(\tau)$ does not necessarily lead to a continuous estimate, which can complicate inference.

There have been several attempts to generalize the notion of quantiles from the real line to higher dimensions, which has been encapsulated in the notion of data depths (R. Y. Liu, Serfling, and Souvaine (2006)). There are a number of notions of data depth, ranging from halfspace depth (Nagy (2025)), simplicial depth (R. Y. Liu (1988)), to the more recent Monge-Kantorovich depth (Chernozhukov, Galichon, et al. (2015)). These notions attempt to define a "central" and "non-central" region for a given distribution, where different notions of depths have different advantages and disadvantages. However, this is not without its perils, as these notions only characterize the distribution under given assumptions, see Nagy (2021) as an example. This work has provided a way to provide a notion of quantiles for functional data, where an overview of recent advancements is given in Gijbels and Nagy (2017).

2.2 Smoothing quantile estimation

The classical quantile estimation problem comes from substituting the empirical cdf, i.e. $\mathbb{F}_n(z) = \frac{1}{n} \sum_{i=1}^n \mathbb{1}\{Z_i \leq z\}$ by the population cdf. However, the substitution means that the risk function is non-smooth in the parameters, which complicates the optimization procedure. This holds true for both quantile estimation and quantile regression estimation. The idea of smoothing this loss function through convolutions originates from Nadaraya (1964) for quantile estimation. The main idea is to smooth the estimate for the empirical cumulative distribution function through a general kernel function K . This function satisfies the assumptions given in Condition 1.

Condition 1 (Kernel function). *A function $K : \mathcal{D} \rightarrow \mathbb{R}$ is defined as a kernel if it is*

A: *The function integrates to 1, i.e. $\int_{\mathcal{D}} K(z) dz = 1$.*

B: *A function K is called an ℓ -th order kernel $\int_{\mathcal{D}} u^j K(u) du = 0$ for all $j = 1, \dots, \ell$ and $\int_{\mathcal{D}} |u|^{\ell+1} K(u) du < \infty$.*

The more recent work by Fernandes, Guerre, and Horta (2021) introduces this notion for regression quantiles with more general kernel functions than Nadaraya (1964). They define the smoothed cdf as $\hat{F}_h(t; \psi) = \int_{-\infty}^t \hat{f}_h(u; \psi) du$ where $\hat{f}_h(u; \psi) = \frac{1}{n} \sum_{i=1}^n \frac{1}{h} K\left(\frac{u - (Z_i - \psi)}{h}\right)$ represents the kernel estimate for the pdf. For the classical quantile estimation problem, this gives the smoothed empirical risk function

$$\hat{R}(\psi; \tau) = \int \rho_{\tau}(u) d\hat{F}_h(u; \psi) = \int \rho_{\tau}(u) \hat{f}_h(u; \psi) du. \quad (3)$$

This can be seen as a convolution that smooths the empirical risk function from a piecewise linear function to a twice differentiable function, simplifying the optimization procedure at the expense of technical cost.

There are a number of different ways to smooth the risk function of the quantile regression estimation problem. First, the function itself can be smoothed, such as in the case of the expectile, which utilizes the square of the difference instead of the absolute value (Newey and Powell (1987)). This is the L^2 norm as opposed to the L^1 norm. This reduces the complexity of the problem, but provides a different interpretation and often requires stricter moment assumptions, reducing the applicability. Second, the indicator term in the loss function of ρ_τ can be smoothed using a kernel function, which provides differentiability to the risk, as in (Horowitz (1998)). However, the covariance structure remains dependent on the probability density function. Hence, they propose a bootstrap procedure to estimate the covariance associated with the estimator. This procedure is similarly utilized in the smoothed estimating equations of Kaplan and Y. Sun (2017) for instrumental variable regression. Third, Gao, Zheng, and Zhu (2025) utilizes a geometric approach to smooth the quantile loss function. They utilize the multiquad function construction to create an infinitely divisible smooth objective function. Their approach works well analytically, but does induce bias without letting the hyperparameter shrink to zero. Fourth, Hutson (2026) smooths the function by approximating the absolute value by a sigmoidal function. This works well in simulations and in their application, but the theoretical results do not seem to be present. Fifth, the errors can be smoothed using a kernel loss function as in Fernandes, Guerre, and Horta (2021). This does not affect the loss function itself, where the empirical distribution is replaced by a smoothed version using kernel convolution smoothing. This method carries the most promise, as it has lower mean squared error than the traditional quantile estimator, a more accurate Bahadur-Kiefer representation, a simplified inference procedure, and shows a theoretical improvement over the work of Horowitz (1998). We focus on the latter smoothing procedure.

This smoothed estimator boosts improved performance over the classical quantile estimator and the Horowitz smoothing procedure. However, the literature has remained relatively sparse on this particular type of smoothing. The framework has been applied to conditional quantile inference in distributed systems in Jiang and Yu (2021). They simplify the objective further using a Taylor approximation of the score to ensure the dependence on the unknown parameter does not make the problem infeasible. The theory has been extended to high-dimensional quantile regression in Tan, Wang, and Zhou (2022). They develop an algorithm that combines the convolution smoothing and concave regularization of the loss function to solve the problem of high-dimensional quantile regression while reducing the bias of the estimators based on the non-smoothed loss function. They show the convergence for the Lasso penalty and that the concave regularization induces a strong oracle property in the estimator. Another addition to the literature considers the Bayesian perspective on smoothing quantile regression by B. Liu, K. Li, and Pang (2025). They provide an alternative to the Bayesian methodology of utilizing an asymmetric Laplacian distribution, which corresponds to the check-loss function, by introducing a new probability density function corresponding to the smooth quantile loss function. They simplify the Markov Chain Monte Carlo inference procedure by utilizing Hamiltonian Monte Carlo, possible due to the smoothness of the new loss function. This paradigm has been applied to functional quantile regression in H. Liu

et al. (2025), which build on the classical theory for functional quantile regression by smoothing the errors in the functional regression. The smoothed loss function has recently been picked up by the machine learning literature to establish ConquerNet (Luo, F. Wu, and W. Wu, 2026), where they train a neural network on the smoothed loss function.

2.3 Extreme value Theory

The assumption of *regular variation* provides the basis for non-degenerate limits of the distribution of maxima within extreme value theory. We define the classical second order regular variation, which controls the rate of convergence towards the limit. We have the following central condition on regularity of the tails

Condition 2. We assume that there exists a function $a : \mathbb{R} \rightarrow \mathbb{R}_{>0}$ and $\gamma \in \mathbb{R}$ such that

$$\lim_{t \rightarrow \infty} \frac{U(tx) - U(t)}{a(t)} = \frac{x^\gamma - 1}{\gamma}, \quad (4)$$

where $U(t) = F^\leftarrow(1 - 1/t)$, for $F^\leftarrow$ as the right-inverse of F .

We denote γ as the *extreme value index*. For the purposes of statistical inference, we can utilize this condition for the Peak-Over-Threshold (POT), or the Block-Maxima paradigm. The POT paradigm selects all values above a high threshold, which often used to estimate the tail index parameter. The selection above a high threshold ensures that the tail follows an, approximately, Generalized Pareto Distribution (GPD), i.e.

$$\lim_{t \rightarrow \infty} \mathbb{P} \left(\frac{Z - U(t)}{a(t)} > z | Z > U(t) \right) = (1 + \gamma z)^{-\frac{1}{\gamma}} =: D_\gamma(z), \quad (5)$$

see Theorem 1.1.6 of De Haan and Ferreira (2006) for details. Alternatively, the block-maxima method divides the data up into regular blocks, and selects the maxima within those blocks. It can be shown that the point process associated with this method converges to a Poisson Point Process (PPP) with the intensity measure dependent on the γ variable. For this work, we mainly focus on the POT paradigm.

There is a wide array of distributions that satisfy this condition. In general, these can be classified into three Domains of Attraction (DoA), where their tails are attracted to a particular form. We have the case of $\gamma > 0$, which is known as the Fréchet DoA. This includes a wide array of heavy-tailed distributions, such as Pareto, Cauchy, student-t and the Burr distributions. These distributions have a simple scaling, where $a(t) = U(t)$ suffices for (5). These distributions always have an unbounded domain. The case of $\gamma = 0$ is known as the Gumbel domain of attraction. This is the domain of light-tailed distributions, such as the Gaussian, exponential, and other distributions. These can either have infinite or finite endpoints. The scaling is more complicated, but does have an explicit form, as denoted in Lemma B.2.18 of Haan and Ferreira (2006). This domain of attraction often provides the most difficulty in inference due to the slow convergence to their DoA. The case of $\gamma < 0$ is known as the Weibull DoA. This DoA includes distributions such as

the Weibull and uniform. These distributions always have a finite endpoint, which is dependent on the extreme value parameter γ . This diversity of distributions suggests that the extreme value condition is relatively weak, as it allows for flexible modeling of the tail behavior. We do note that there are limits to this, as one can show that the set of discrete distributions is precluded from this set.

This extreme value condition can be further refined to capture not only the convergence of high quantiles to a specific form, but additionally to define the rate of convergence towards the tail, which is dubbed the second order extreme value condition. This is encapsulated in Condition 3.

Condition 3 (Second Order Condition). *In addition to (4), we assume that there exists a function A and a second order index $\rho \leq 0$, which is eventually positive or negative, such that*

$$\lim_{t \rightarrow \infty} \frac{\frac{U(tx) - U(t)}{a(t)} - \frac{x^\gamma - 1}{\gamma}}{A(t)} = \Psi_{\gamma, \rho}(x), \quad (6)$$

where the form $\Psi_{\gamma, \rho}(x)$ is given in Theorem B.3.10 of De Haan and Ferreira (2006).

This condition is more stringent than the extreme value condition, as, for example, the Generalized Pareto Distribution does not admit such a rate. This condition is often utilized to establish asymptotic normality for different estimators, such as in Ferreira and De Haan (2015). This condition restricts the set of distributions slightly, as the Generalized Pareto distribution or the exponential distribution satisfy (4) but not (6). A more detailed overview is given in Section B.3 of Haan and Ferreira (2006).

The traditional literature on empirical process theory is exceptionally powerful in establishing convergence results for different types of statistics, see e.g. Van der Vaart (2000). This holds true in the extreme value theory literature as well, where the *tail empirical process* has played a vital role in advancing the theory of estimation within the field. Hence, if we have results for this process, then we can proceed accordingly. To this end, we consider Theorem 5.1.2 from Haan and Ferreira (2006). This will provide the basis for further investigation.

Theorem 2.1 (Theorem 5.1.2 of Haan and Ferreira (2006)). *Let $\{X_i\}_{i=1}^n$ be i.i.d. random variables with distribution function F . Let $\mathbb{F}_n$ be the empirical distribution function, based on $X_1, X_2, \dots, X_n$. Suppose that U , satisfies the second order condition given in 3 with $\gamma \in \mathbb{R}$ and $\rho \leq 0$. Let $k := k(n)$ be a sequence of integers such that $k \rightarrow \infty$ and $\sqrt{k}A_0\left(\frac{n}{k}\right)$ is bounded as $n \rightarrow \infty$. Then, the underlying sample space can be enlarged to include a sequence of Brownian Motion W_n such that for all $x_0 > -\frac{1}{\gamma \vee 0}$:*

1. *If not $\gamma = \rho = 0$, then as $n \rightarrow \infty$*

$$\sup_{x_0 \leq x \leq \frac{1}{(-\gamma) \vee 0}} D_\gamma(x)^{-1/2+\varepsilon} \left| \sqrt{k} \left[\frac{n}{k} \left(1 - \mathbb{F}_n \left(a_0 \left(\frac{n}{k} \right) x + U \left(\frac{n}{k} \right) \right) \right) - D_\gamma(x) \right] \right. \\ \left. - W_n(D_\gamma(x)) - \sqrt{k}A_0 \left(\frac{n}{k} \right) D'_\gamma(x) \bar{\Psi}_{\gamma, \rho}(D_\gamma(x)) \right| = o_P(1).$$

2. If $\gamma = \rho = 0$, then as $n \rightarrow \infty$ and for all $\delta > 0$,

$$\sup_{x_0 \leq x < \infty} \max(1, x)^\delta \left| \sqrt{k} \left[\frac{n}{k} \left(1 - \mathbb{F}_n \left(a_0 \left(\frac{n}{k} \right) x + U \left(\frac{n}{k} \right) \right) \right) - e^{-x} \right] - W_n(e^{-x}) - \sqrt{k} A_0 \left(\frac{n}{k} \right) \frac{x^2}{e^{-x}} \right| = o_P(1).$$

This is the main theorem utilized for the i.i.d case, and will be useful for establishing our results w.r.t consistency, asymptotic normality, and second order consistency. The tail empirical process is very closely related to the distribution of the order statistics for extremes. In particular, one can construct the tail quantile process by utilizing two results. First, the weak convergence of Theorem 2.1 allows one to invoke a Skorohod construction, see Theorem A.0.1 of Haan and Ferreira (2006), to convert the weak convergence into almost sure convergence. Second, Vervaat's Lemma, see Lemma A.0.2 of Haan and Ferreira (2006), can be utilized to invert the a.s. limit to achieve the asymptotics for the quantile tail process. The details can be found in Theorem 2.4.2 of Haan and Ferreira (2006).

The work of Gnedenko (1943) provides the first characterization of the limit distribution of the high-quantile estimator under extreme value conditions for extremal order statistics, i.e. the quantile τ_n such that $n\tau_n \rightarrow \tau^*$ as $n \rightarrow \infty$. This result holds for i.i.d sequences of random variables, where they showed that

$$\sqrt{n\tau_n} \frac{\hat{U} \left(\frac{1}{\tau_n} \right) - U \left(\frac{1}{\tau_n} \right)}{a_0 \left(\frac{1}{\tau_n} \right)} \xrightarrow{d} N(0, 1).$$

This was later generalized to include dependency in the data. In particular, Meyer (1973) provides a generalization to stationary processes with low probabilities of extreme events clustering using a Poisson point process approach. The book of Leadbetter, Lindgren, and Rootzén (2012) provides an overview of the literature on including dependency between observations for extremes for such processes.

The main relevance for intermediate quantile for extremes as opposed to the extreme quantiles arises from the need to choose a relatively high value in the POT framework. The intermediate quantile functions as a good choice for this. If we consider Condition 3, then we see that we require knowledge of $U(t), a_0(t)$ for this result to work, similarly for Theorem 2.1. Then, most estimators for γ rely on a good approximation of $U(t)$ at a certain level t to work well. A natural bias-variance trade-off occurs here, as we would like to be as deep in the tail as possible to minimize the bias, while utilizing as much observations as possible to minimize the variance. Hence, we would like to choose our level k and our approximation for U smartly, as to minimize this trade-off.

3 Methodology

This section sets out our theoretical results for the smoothed intermediate quantile estimation problem. We prove global consistency and the asymptotic normality for the intermediate quantile estimation under the correct scaling. Furthermore, we prove that the second order derivative provides a consistent estimator for the correct scaling, which provides an improvement over classical approaches.

Throughout the section, we utilize the concept of a higher order kernel introduced in Condition 1.B. However, these kernels are rather difficult to bound, as opposed to the second order kernels. This class is easier to work with, as it can often be reduced to the class of probability density functions. The advantage of these second order kernels is that they have known tail behaviors, i.e. how the function behaves over large values. The tail behavior will be rather important, as this dictates the degree to which we smooth the estimate. To this end, we impose the following condition on the class of kernels.

Condition 4 (sub-Gaussian absolute tail bounds). *For a given kernel $K : \mathbb{R} \rightarrow \mathbb{R}$, we suppose that there exists constants $C_K, c_K > 0$ and $t_0 > 0$ such that*

$$|K(t)| \leq C_K e^{-c_K t^2}, \quad \forall |t| \geq t_0.$$

Examples of classes of kernels that satisfy this bound are the higher order Epanichekov, uniform, or Gaussian kernels Hansen (2005) and the dilitation kernels with a bounded underlying function Kerkycharian, Lepski, and Picard (2001). This condition ensures that the integral of this kernel is appropriately bounded. This can equivalently be formulated as a condition on the normalized absolute kernel, which is a probability density function by definition. This condition can likely be relaxed to allow for more general tail behaviors, which would additionally have a closer link to the order of the kernel. We discuss this further in Section 6.

The smoothness of the objective function induced through the convolution means that we can differentiate. The derivative of (3) can be written as

$$\begin{aligned} \frac{n}{k} \frac{\partial \hat{R}(\psi; 1 - k/n)}{\partial \psi} &= \frac{1}{n} \sum_{i=1}^n \int_0^\infty \frac{n}{k} K_h(u - (Z_i - \psi)) du - 1 \\ &= \frac{1}{n} \sum_{i=1}^n \int_0^\infty \int_{-\infty}^\infty \frac{n}{k} \frac{1}{h} K'_h(u - (z - \psi)) \mathbb{1}\{Z_i > z\} dz du - 1 \\ &= \frac{1}{n} \sum_{i=1}^n \int_{-\infty}^\infty \int_0^\infty \frac{n}{k} \frac{1}{h} K'_h(u - (z - \psi)) \mathbb{1}\{Z_i > z\} du dz - 1 \\ &= \int_{-\infty}^\infty \frac{n}{k} (1 - \mathbb{F}_n)(\psi - z) K_h(z) dz - 1, \end{aligned}$$

where k is the intermediate sequence such that $\frac{k}{n} \rightarrow 0$ and $k \rightarrow \infty$ as $n \rightarrow \infty$. This transforms our original M-estimation problem into a Z-estimation problem see (Van der Vaart, 2000). Hence, in a similar vein to Lemma 5.10 of Van der Vaart (2000), we want to establish that this function is non-increasing for all $\psi \in \mathbb{R}$ and that it is decreasing around the true value $\psi = U\left(\frac{n}{k}\right)$. To achieve the consistency of this

estimator, we require that $h \rightarrow 0$ as $n \rightarrow \infty$ at a certain rate. This provides the following consistency result.

Lemma 3.1 (Consistency). *Suppose the second order extreme value condition (6) is satisfied. Assume Condition 4 holds and that $h_n^{-1} = o(|\log(k/n)|^{-1/2})$ as $n \rightarrow \infty$ for $h_n = \frac{h}{a_0(n/k)}$. Then, we have that there exists $\hat{\psi}(\frac{n}{k})$ which is the solution to*

$$\int_{-\infty}^{\infty} \frac{n}{k} (1 - \mathbb{F}_n)(\psi - z) K_h(z) dz - 1 = 0,$$

such that $\frac{\hat{\psi}(\frac{n}{k}) - U(\frac{n}{k})}{a_0(\frac{n}{k})} \xrightarrow{P} 0$ as $n \rightarrow \infty$.

Proof of Lemma 3.1. By the smoothness of the convolution, we note that the monotonicity of the first order derivative of the risk function (see (3)) holds across all of $\mathbb{R}$, immediately for unbounded kernels. For bounded kernels with domain $[-b, b]$, we note that the function is monotonic on $[Z_{n,1} - bh_n, Z_{n,n} + bh_n]$. Therefore, as $Z_{n,1}$ and $Z_{n,n}$ converge to the left- and right-endpoints of the distribution, respectively, this holds with high probability. This ensures the existence of the estimator. Thus, we mainly focus on the second requirement, which ensures uniqueness in the limit. We rewrite the integral to achieve the correct scaling, i.e.

$$\int_{-\infty}^{\infty} \frac{n}{k} (1 - \mathbb{F}_n) \left(a_0\left(\frac{n}{k}\right) \left(\frac{\psi - U(\frac{n}{k})}{a_0(\frac{n}{k})} - z \right) + U\left(\frac{n}{k}\right) \right) K_{\frac{h}{a_0(\frac{n}{k})}}(z) dz - 1.$$

Then, for $\phi_n := \frac{\psi - U(\frac{n}{k})}{a_0(\frac{n}{k})} \in B_\varepsilon(0)$ for $\varepsilon > 0$, we can utilize Lemma A.1 to be able to write that

$$\begin{aligned} & \int_{-\infty}^{\infty} \frac{n}{k} (1 - \mathbb{F}_n) \left(a_0\left(\frac{n}{k}\right) (\phi_n - z) + U\left(\frac{n}{k}\right) \right) K_{\frac{h}{a_0(\frac{n}{k})}}(z) dz - 1 \\ &= D_\gamma(\phi_n) - 1 + o_P\left(\frac{1}{\sqrt{k}}\right), \end{aligned}$$

where $D_\gamma(\phi)$ is the survival function of the GPD, see (5), and the remainder term is uniform over all $\phi \in B_\varepsilon(0)$. Hence, the function is strictly decreasing in a neighborhood around the true value of $\phi = 0$ with high-probability. Thus, the estimator $\hat{\psi}(\frac{n}{k})$ satisfies the first order condition by construction,

$$\int_{-\infty}^{\infty} \frac{n}{k} (1 - \mathbb{F}_n)(\psi - z) K_h(z) dz - 1 = 0,$$

we see that $\frac{\hat{\psi}(\frac{n}{k}) - U(\frac{n}{k})}{a_0(\frac{n}{k})} \xrightarrow{P} 0$ as $n \rightarrow \infty$. □

The consistency and asymptotic normality requires knowledge of the $a_0(n/k)$ function. The current literature solves this through approximation using the estimated quantiles, see Chernozhukov, Fernández-Val, and Kaji (2016). We show

that the second order derivative of the risk function provides a novel unbiased estimator for the scale function. This estimator is straightforward to compute, arises naturally, and does not require bootstrap results to be shown. This extends to the intermediate quantile estimation case, as shown in Theorem 3.3. However, it extends more generally, providing another estimator for the scaling function $a_0\left(\frac{n}{k}\right)$, which does not inflate the variance.

Lemma 3.2. *Suppose that a general estimator $\hat{U}\left(\frac{n}{k}\right)$ of $U\left(\frac{n}{k}\right)$ is consistent in the sense that $\frac{\hat{U}\left(\frac{n}{k}\right) - U\left(\frac{n}{k}\right)}{a_0\left(\frac{n}{k}\right)} \xrightarrow{P} 0$ as $n \rightarrow \infty$. We assume Condition 4 and that $h_n^{-1} = o\left(|\log(k/n)|^{-1/2}\right)$ as $n \rightarrow \infty$ for $h_n = \frac{h}{a_0(n/k)}$. Then, we have that*

$$a_0\left(\frac{n}{k}\right) \frac{1}{n} \sum_{i=1}^n \frac{n}{k} K_h\left(\hat{U}\left(\frac{n}{k}\right) - Z_i\right) \xrightarrow{P} 1.$$

Thus, we can replace the unknown scaling coefficient $a_0\left(\frac{n}{k}\right)$ in the consistency and asymptotic normality statements by a simple estimator. This will allow for cleaner asymptotics and inference through smoothing.

Proof of Lemma 3.2. We can rewrite the above expression as

$$a_0\left(\frac{n}{k}\right) \frac{1}{n} \sum_{i=1}^n \frac{n}{k} K_h\left(\hat{U}\left(\frac{n}{k}\right) - Z_i\right) = \frac{1}{n} \sum_{i=1}^n \frac{n}{k} K_{h_n}\left(\frac{\hat{U}\left(\frac{n}{k}\right) - U\left(\frac{n}{k}\right)}{a_0\left(\frac{n}{k}\right)} - \frac{Z_i - U\left(\frac{n}{k}\right)}{a_0\left(\frac{n}{k}\right)}\right).$$

We can decompose the difference as

$$\begin{aligned} & \frac{1}{n} \sum_{i=1}^n \frac{n}{k} K_{h_n}\left(\frac{\hat{U}\left(\frac{n}{k}\right) - U\left(\frac{n}{k}\right)}{a_0\left(\frac{n}{k}\right)} - \frac{Z_i - U\left(\frac{n}{k}\right)}{a_0\left(\frac{n}{k}\right)}\right) - 1 \\ &= \frac{1}{n} \sum_{i=1}^n \frac{n}{k} K_{h_n}\left(\frac{\hat{U}\left(\frac{n}{k}\right) - U\left(\frac{n}{k}\right)}{a_0\left(\frac{n}{k}\right)} - \frac{Z_i - U\left(\frac{n}{k}\right)}{a_0\left(\frac{n}{k}\right)}\right) - D_\gamma\left(\frac{\hat{U}\left(\frac{n}{k}\right) - U\left(\frac{n}{k}\right)}{a_0\left(\frac{n}{k}\right)}\right) \\ & \quad + D_\gamma\left(\frac{\hat{U}\left(\frac{n}{k}\right) - U\left(\frac{n}{k}\right)}{a_0\left(\frac{n}{k}\right)}\right) - 1 \\ &= I_1 + I_2. \end{aligned}$$

For I_1 and $\varepsilon > 0$, we condition on the event that $\varepsilon > 0$ that $\left\{\left|\frac{\hat{U}\left(\frac{n}{k}\right) - U\left(\frac{n}{k}\right)}{a_0\left(\frac{n}{k}\right)}\right| \leq \varepsilon\right\}$.

Then, by Lemma A.3, if we set $\varepsilon \leq r < \frac{1}{|\gamma|}$, then we have that $I_1 \xrightarrow{P} 0$.

For I_2 , we can use the continuous mapping theorem to write that $I_2 \xrightarrow{P} D_\gamma(0) - 1 = 0$. $\square$

The following result characterizes the asymptotic behavior of the estimator.

Theorem 3.3. *Suppose the second order extreme value condition holds with $\sqrt{k}A_0\left(\frac{n}{k}\right) \rightarrow \lambda \in \mathbb{R}$ as $n \rightarrow \infty$. Further, under Conditions 4 and 1.B, assume that $h_n^{-1} =$*

$o\left(|\log(\sqrt{k}/n)|^{-1/2}\right)$ and $h_n^{\ell+1}\sqrt{k} = o(1)$ as $n \rightarrow \infty$ for $h_n = \frac{h}{a_0(n/k)}$. Then, we have that the smoothed quantile estimator has the asymptotic limit

$$\sqrt{k} \frac{\left[\hat{\psi}\left(\frac{n}{k}\right) - U\left(\frac{n}{k}\right)\right]}{a_0\left(\frac{n}{k}\right)} \xrightarrow{d} N(-\lambda \bar{\Psi}_{\gamma, \rho}(1), 1), \quad n \rightarrow \infty. \quad (7)$$

Proof of Theorem 3.3. The asymptotic normality of the estimator can be established using Corollary 5.58 from Van der Vaart (2000), which is a consequence of the Argmax Theorem. For $\nu \in B_{r_n}(0)$ where $r_n = O(k^{\delta_H})$ for $0 < \delta_H < 1/2$, we define

$$\nu = \sqrt{k} \frac{\psi - U\left(\frac{n}{k}\right)}{a_0\left(\frac{n}{k}\right)} \iff \psi = \frac{\nu}{\sqrt{k}} a_0\left(\frac{n}{k}\right) + U\left(\frac{n}{k}\right).$$

Then, we can define

$$\begin{aligned} M_n(\nu) &= \frac{1}{a_0\left(\frac{n}{k}\right)} \sum_{i=1}^n \int_{-\infty}^{\infty} \rho_{1-k/n}(u) K_h\left(u - \left(Z_i - \frac{\nu}{\sqrt{k}} a_0\left(\frac{n}{k}\right) - U\left(\frac{n}{k}\right)\right)\right) du \\ &= \frac{n}{a_0\left(\frac{n}{k}\right)} \hat{R}\left(\frac{\nu}{\sqrt{k}} a_0\left(\frac{n}{k}\right) + U\left(\frac{n}{k}\right); 1 - \frac{k}{n}\right). \end{aligned}$$

Then, through a second order Taylor expansion with remainder term, there exists $\tau \in [0, 1]$ such that

$$\begin{aligned} M_n(\nu) - M_n(0) &= \nu \frac{\partial M_n(0)}{\partial \nu} + \frac{1}{2} \nu^2 \frac{\partial^2 M_n(\nu\tau)}{\partial \nu^2} \\ &= \nu \frac{n}{\sqrt{k}} \frac{\partial \hat{R}}{\partial \psi}\left(U\left(\frac{n}{k}\right); 1 - \frac{k}{n}\right) + \frac{\nu^2}{2} a_0\left(\frac{n}{k}\right) \frac{n}{k} \frac{\partial^2 \hat{R}}{\partial \psi^2}\left(\frac{\nu}{\sqrt{k}} a_0\left(\frac{n}{k}\right) + U\left(\frac{n}{k}\right); 1 - \frac{k}{n}\right) \\ &= \nu \sqrt{k} \left(\frac{n}{k} \int_{-\infty}^{\infty} (1 - \mathbb{F}_n)\left(U\left(\frac{n}{k}\right) - z\right) K_h(z) dz - 1\right) \\ &\quad + \frac{\nu^2}{2} a_0\left(\frac{n}{k}\right) \frac{1}{n} \sum_{i=1}^n \frac{n}{k} K_h\left(\frac{\nu}{\sqrt{k}} a_0\left(\frac{n}{k}\right) + U\left(\frac{n}{k}\right) - Z_i\right) \end{aligned}$$

Then, we have to ensure the first order and second order derivatives converge appropriately. This allows us to divide the proof up into three parts.

1. For the first order derivative, we can rewrite the term as

$$\begin{aligned} &\sqrt{k} \left(\frac{n}{k} \int_{-\infty}^{\infty} (1 - \mathbb{F}_n)\left(U\left(\frac{n}{k}\right) - z\right) K_h(z) dz - 1\right) \\ &= \sqrt{k} \left(\frac{n}{k} \int_{-\infty}^{\infty} (1 - \mathbb{F}_n)\left(U\left(\frac{n}{k}\right) - z a_0\left(\frac{n}{k}\right)\right) K_{h_n}(z) dz - 1\right) \\ &= J \end{aligned}$$

Then, utilizing Lemma A.2, we have that this term converges in distribution to

$$J \xrightarrow{d} N(0, 1) + \lambda \bar{\Psi}_{\gamma, \rho}(1),$$

where the second term represent the bias from the second order approximation.

2. For the second order term up to a fraction $\frac{\nu^2}{2}$, we can rewrite the term as

$$\begin{aligned} & a_0\left(\frac{n}{k}\right) \frac{1}{n} \sum_{i=1}^n \frac{n}{k} K_h \left(\frac{\nu}{\sqrt{k}} a_0\left(\frac{n}{k}\right) + U\left(\frac{n}{k}\right) - Z_i \right) \\ &= \frac{1}{n} \sum_{i=1}^n \frac{n}{k} K_{h_n} \left(\frac{\nu}{\sqrt{k}} + \frac{U\left(\frac{n}{k}\right) - Z_i}{a_0\left(\frac{n}{k}\right)} \right), \end{aligned}$$

where we note that $\frac{\nu}{\sqrt{k}} = o(1)$ as $n \rightarrow \infty$. Then, we can apply Lemma A.3, by which the above converges to $D'_\gamma(0) = 1$.

3. Then, by Corollary 5.58 from Van der Vaart (2000), we can define

$$\overline{M}_n(\nu) := M_n(\nu) - M_n(0), \quad \overline{M}(\nu) := \nu(\lambda \overline{\Psi}_{\gamma, \rho}(1) + Z) + \frac{\nu^2}{2},$$

for $Z \sim N(0, 1)$. By the previous points, we have that $\overline{M}_n(\nu) \xrightarrow{d} \overline{M}(\nu)$ for $\nu \in B_{r_n}(\nu_0)$. Then,

$$\hat{\nu}_n = \arg \max_{\nu \in H_n} \overline{M}_n(\nu) \xrightarrow{d} \arg \max_{\nu \in H} \overline{M}(\nu) = N(-\lambda \overline{\Psi}_{\gamma, \rho}(1), 1),$$

provided that the sequence $\hat{\nu}_n$ is tight. The tightness holds by the consistency result given in Lemma 3.1.

□

The following corollary is a direct consequence of Theorem 3.3 and Lemma 3.2.

Corollary 3.3.1. *Suppose the second order extreme value condition holds with $\sqrt{k}A_0\left(\frac{n}{k}\right) \rightarrow \lambda \in \mathbb{R}$ as $n \rightarrow \infty$. Further, under Conditions 4 and 1.B, assume that $h_n^{-1} = o\left(|\log(\sqrt{k}/n)|^{-1/2}\right)$ and $h_n^{\ell+1}\sqrt{k} = o(1)$ as $n \rightarrow \infty$ for $h_n = \frac{h}{a_0(n/k)}$. Then, we have that the smoothed quantile estimator has the asymptotic limit*

$$\sqrt{k} \left[\hat{\psi}\left(\frac{n}{k}\right) - U\left(\frac{n}{k}\right) \right] \cdot \frac{1}{n} \sum_{i=1}^n \frac{n}{k} K_h \left(\hat{\psi}\left(\frac{n}{k}\right) - Z_i \right) \xrightarrow{d} N(-\lambda \overline{\Psi}_{\gamma, \rho}(1), 1), \quad n \rightarrow \infty. \quad (8)$$

4 Simulations

The main obstacle in the current methodology is that the unknown scaling function $a_0(n/k)$. We require this parameter to effectively define a bandwidth parameter. We investigate the effect of several possible bandwidth estimators based on classical extreme value statistics. We provide validation of the theoretical results and provide simulations that indicate that an estimate of the scaling parameter still allows for an effective bandwidth choice.

We provide results for the distributions given in Table 1. These distributions cover all domains of attraction. The second order parameter ρ_0 for the Burr12 distribution has an explicit form. Hence, we can vary both the self-similarity structure and the rate of convergence to this structure using this distribution.

Domain of Attraction	Distribution	PDF	Parameter(s)
Weibull	Beta	$\frac{z^{\alpha-1}(1-z)^{\beta-1}}{B(\alpha,\beta)}$	$(\alpha, \beta) = \left(1, -\frac{1}{\gamma_0}\right)$
Gumbel	Normal	$\frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}z^2\right)$	-
Fréchet	Student-t	$\frac{\Gamma(\frac{\nu+1}{2})}{\sqrt{\pi\nu}\Gamma(\frac{\nu}{2})} \left(1 + \frac{z^2}{\nu}\right)^{-\frac{\nu+1}{2}}$	$\nu = \frac{1}{\gamma_0}$
Fréchet	Burr12	$cd \frac{z^{c-1}}{(1+x^c)^{d+1}}$	$(c, d) = \left(-\frac{1}{\rho_0}, -\frac{1}{\gamma_0\rho_0}\right)$

Table 1: Overview of distributions considered in the simulation studies with parameter configurations for the extreme value index γ_0 and the second order index ρ_0 .

The theoretical results require that $\frac{a_0(n/k)}{h}$ converges to zero as $n \rightarrow \infty$ for some choice of $h > 0$. We model the bandwidth as

$$h = \bar{h} \cdot f(\{X_{n-i,n}\}_{i=0}^k).$$

We choose functions f such that $\frac{a_0(n/k)}{f(\{X_{n-i,n}\}_{i=0}^k)} \xrightarrow{P} \eta \in [0, \infty)$ as $n \rightarrow \infty$. Then, we can control the rate of convergence through the $\bar{h}$ bandwidth. First, we consider the "naïve" estimator based on the tail empirical quantile process, see Haan and Ferreira (2006), where $f^N(\{X_{n-i,n}\}_{i=0}^k) = X_{n-[k/2],n} - X_{n-k,n}$. We know that $\frac{a_0(n/k)}{X_{n-[k/2],n} - X_{n-k,n}} \xrightarrow{P} \frac{\gamma_0}{(\frac{1}{2})^{\gamma_0} - 1}$, which is a positive constant. Second, we consider the "averaged" estimator of as $f(\{X_{n-i,n}\}_{i=0}^k) = \frac{1}{k(1-\alpha)} \sum_{i=[k\alpha]}^k (X_{n-i,n} - X_{n-k,n})$ for $\alpha > 0$ as a cutoff parameter. We choose this cutoff above zero, as the results for the quantile process can not be guaranteed for the maximum by Theorem 2.4.2 of Haan and Ferreira (2006). By the convergence of the quantile process, we have

$$\begin{aligned} \frac{f^{\text{avg}}(\{X_{n-i,n}\}_{i=0}^k)}{a_0\left(\frac{n}{k}\right)} &= \frac{1}{k(1-\alpha)} \sum_{i=[k\alpha]}^k \frac{(X_{n-i+1,n} - X_{n-k,n})}{a_0\left(\frac{n}{k}\right)} \\ &= \int_{\alpha}^1 \frac{(X_{n-[ku],n} - X_{n-k,n})}{a_0\left(\frac{n}{k}\right)} du \xrightarrow{P} \int_{\alpha}^1 \frac{\left(\frac{1}{u}\right)^{\gamma_0} - 1}{\gamma_0} du. \end{aligned}$$

Finally, we consider the "weighted-average" scale estimator based on the work of Caeiro, Mateus, and Soltane (2021) for the Hill-estimator. We define the weight function $w_a(u) = 1 + a - 4a \cdot u$. Then, we define the estimate

$$\begin{aligned} \frac{f^{WA}(\{X_{n-i,n}\}_{i=0}^k)}{a_0\left(\frac{n}{k}\right)} &= \frac{1}{k(1-\alpha)} \sum_{i=[k\alpha]}^k w_a\left(\frac{i}{k+1}\right) \frac{(X_{n-i+1,n} - X_{n-k,n})}{a_0\left(\frac{n}{k}\right)} \\ &= \int_{\alpha}^1 w_a([ku]) \frac{(X_{n-[ku],n} - X_{n-k,n})}{a_0\left(\frac{n}{k}\right)} du \xrightarrow{P} \int_{\alpha}^1 w_a(u) \frac{\left(\frac{1}{u}\right)^{\gamma_0} - 1}{\gamma_0} du. \end{aligned}$$

Then, it remains to control the bandwidth $\bar{h}$ based on the tail behavior of the kernel.

Throughout the simulations, we compute the test-statistic

$$T = \sqrt{k} [\hat{\psi}(n/k) - U(n/k)] \cdot \frac{1}{n} \sum_{i=1}^n \frac{n}{k} K_h(\hat{\psi}(n/k) - Z_i)$$

across 100 independent replications at a sample size of $n = 2000$. We consider the set of levels $k/n \in \{0.05, 0.10, 0.15, 0.20, 0.25\}$, and the set of bandwidth powers $\bar{h} \in \{0.10, 0.25, 0.50, 0.75, 1.00\}$. Theorem 3.3 predicts $T \rightarrow N(-\lambda\bar{\Psi}_{\gamma,\rho}(1), 1)$; we track both the bias $\mathbb{E}[T] \approx -\lambda\bar{\Psi}_{\gamma,\rho}(1)$ and whether $\text{Var}(T) = 1$, the latter guaranteed by Lemma 3.2 via the second-order risk derivative. We present the results in three parts. First, Section 4.1 provides a general investigation of the naïve and truncated estimators. Second, Section 4.2 investigates the behaviour of the untruncated mean-based estimators bandwidth breaks down. Third, Section 4.3 investigates the effect of the asymptotic bias induced by the extreme-value approximation.

4.1 Theoretical Verification

This section verifies the theoretical results by comparing the naïve estimator against the two truncated mean-based estimators. We denote the average truncated estimator and the weighted average truncated estimator as Avg. (trunc.) and Wtd. avg. (trunc.). We utilize the hyperparameter choice $\alpha = 0.8$ throughout to put more weight on lower order statistics. This is to provide robustness to our estimators. We utilize the Gaussian kernel throughout. This kernel satisfies Condition 4, and comes as close as possible.

By Theorem 3.3, we require that the bandwidth satisfies $h_n/a_0(n/k) \rightarrow 0$. We start by verifying this condition for our estimators against the oracle estimator. We compute the oracle estimator $a_0(n/k) = (k/n)/f(F^{-1}(1 - k/n))$ is computed analytically for each distribution. Figure 3 shows the ratio $h_n/a_0(n/k)$ at $k/n = 0.15$; lines and shading give the mean and 80% simulation interval over 100 replications.

The choices for the bandwidths indicate that the bandwidth condition is satisfied. Across all three domains of attraction, the three estimators seem to satisfy the condition of converging faster to zero than the oracle. There is a clear ordering to the convergences, where across all three domains of attraction, we see that the naïve estimator has a higher bandwidth than the average and weighted average estimators. This is likely due to the fact that the naïve estimator chooses slightly larger order statistics than the average and weighted average choices. We note that

this condition persists to the Fréchet domain of attraction with the heavy-tailed student-t distribution. This suggests that the stability of the bandwidths ensured in the theory can be retained.

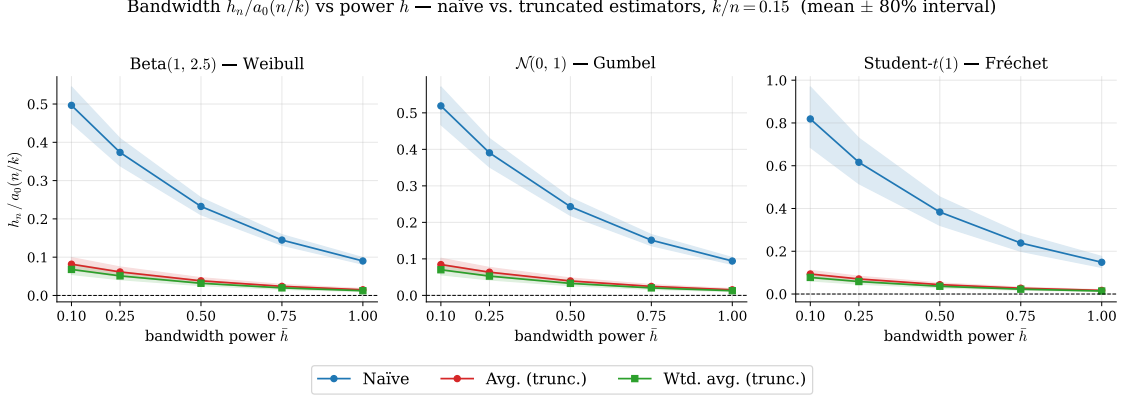


Figure 3: Ratio $h_n/a_0(n/k)$ as a function of bandwidth power $\bar{h}$ at $k/n = 0.15$, for the Beta (Weibull), Normal (Gumbel), and Student- $t(1)$ (Fréchet) distributions, comparing the naïve estimator to its truncated average and weighted-average counterparts. Lines show the mean over 100 simulations; shading covers the 10th–90th percentile interval attained from simulations. A ratio below one indicates the bandwidth condition of Theorem 3.3 is satisfied.

We consider the behaviour of the bias of the statistic T summarized in Table 2. The table displays the estimated bias, standard deviation, and the root-mean-squared-error (RMSE) attained from the simulations. We see distinct patterns emerge across the domains of attraction. The Beta distribution in the Weibull domain of attraction has a relatively low bias and variance overall, indicating that the method works well. The Gumbel domain has a higher RMSE, which is mainly dominated by the bias that sets in. The intuition here is that the Gaussian distribution converges rather slowly to the tail limiting distribution, at approximately rate $\sqrt{n}$. Hence, there is bound to be a bias that affects the estimation procedure. The student-t distribution is clearly the most difficult to estimate, where the simulations indicate a clear bias and an elevated standard error for the estimators. The elevated bias and standard errors can be explained by the heavy-tailed property of the distribution, where large deviations from the true quantile of interest make it more difficult to estimate. This is particularly pertinent as the Gaussian kernel gives weight to the errors of all observations w.r.t the quantile, not only the closest one such as in the classical approach. In addition, there is a clear trend across the estimators, where the average and weighted average estimators, which put higher importance on lower order statistics, seem to perform better across both the bias and the standard deviation. This suggests that additional information may contribute to improved estimation.

Distribution	Scale estimator	$\widehat{\mathbb{E}[T]}$	$\sqrt{\widehat{\text{Var}(T)}}$	RMSE
Beta(1, 2.5)	Naïve	0.001	0.870	0.866
	Avg. (trunc.)	-0.001	0.887	0.883
	Wtd. avg. (trunc.)	0.001	0.893	0.889
$\mathcal{N}(0, 1)$	Naïve	-0.144	0.905	0.912
	Avg. (trunc.)	-0.145	0.949	0.956
	Wtd. avg. (trunc.)	-0.145	0.961	0.967
Student- $t(1)$	Naïve	0.858	1.707	1.903
	Avg. (trunc.)	0.610	1.426	1.545
	Wtd. avg. (trunc.)	0.399	1.241	1.297

Table 2: Empirical $\widehat{\mathbb{E}[T]}$, $\widehat{\text{Var}(T)}$, and $\text{RMSE} = \sqrt{\widehat{\mathbb{E}[T^2]}}$ at $\bar{h} = 0.5$, $k/n = 0.15$, $n = 2000$, 100 replications, for the naïve estimator and its two truncated mean-based competitors. $\widehat{\mathbb{E}[T]}$ estimates $-\lambda\bar{\Psi}_{\gamma,\rho}(1)$; the theoretical target for $\widehat{\text{Var}(T)}$ is one.

We consider the bias and the standard deviation across the bandwidth powers in Figures 4 and 5. Both the bias and the standard deviation seem to follow very different patterns across domains of attraction. We treat each in turn. For the Beta distribution in the Weibull domain of attraction, we see that a higher bandwidth introduces both higher bias and standard deviation for average and weighted average estimators. For the naïve estimator, the standard deviation bottoms out around $h = 0.50$, while the bias stabilizes around the same value. This likely has to do with the slightly higher order statistics that are retained in the naïve estimator, where a slightly higher bandwidth stabilizes the estimator.

For the Gumbel domain of attraction with the Gaussian distribution, we see that the bias seems to remain relatively constant across power, while the standard deviation seems to increase monotonically for higher bandwidths. This likely has to do with the approximation rate of the Gaussian distribution, where order statistics carry relatively little information on the limiting distribution.

For the Fréchet domain of attraction with the student- t distribution, the bias and standard deviation follow a similar pattern. The average and weighted average estimators are relatively stable across bandwidth values, while the naïve estimator shows a clear decrease for higher bandwidth values. This likely has to do with a practical reason, as the deviations from the estimated quantile lead to the kernel estimator of the second order derivative to be relatively sparsely weighted. Hence, a higher bandwidth can stabilize this procedure as opposed to a lower bandwidth which can cause more numerical problems.

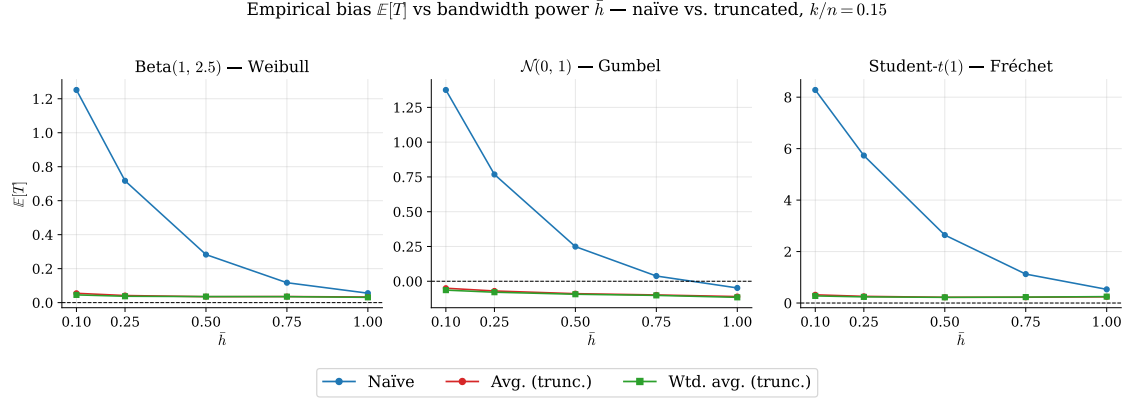


Figure 4: Empirical mean $\widehat{\mathbb{E}[T]}$ as a function of bandwidth power $\bar{h}$ at $k/n = 0.15$. The dashed line marks zero. For Beta and Normal all estimators sit near zero regardless of power. For Student- t , the truncated estimators sit below the naïve estimator throughout the power grid.

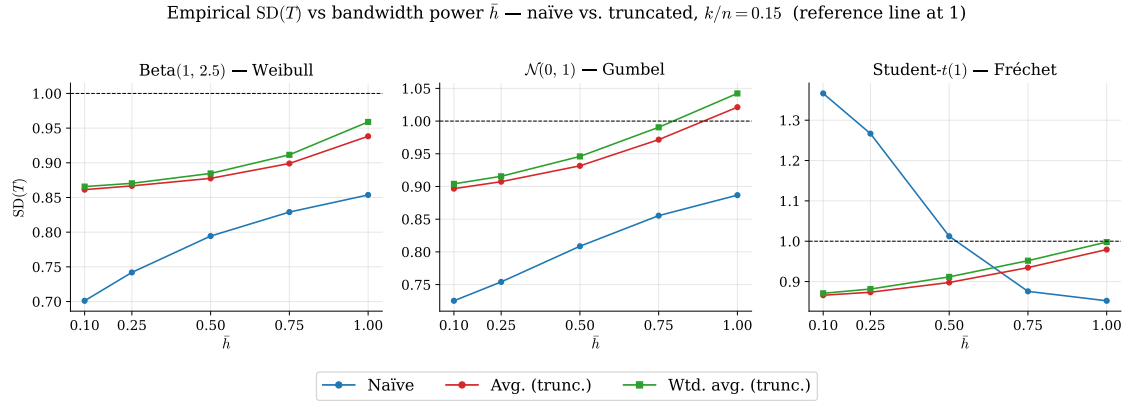


Figure 5: Empirical standard deviation $\widehat{\text{Var}(T)}$ as a function of bandwidth power $\bar{h}$ at $k/n = 0.15$. The dashed line marks the theoretical target of one. For Beta and Normal, $\widehat{\text{Var}(T)}$ is flat across powers and slightly larger for the truncated estimators. For Student- t , the truncated estimators achieve the lowest $\widehat{\text{Var}(T)}$ of the three.

We have mainly considered the consistency aspect of the estimator through the RMSE and the associated bias and variance. However, we established asymptotic normality of the estimator in question, which allows us to perform inference. We verify these claims by comparing the T -statistic to a histograms and QQ-plots in Figures 6 and 7 comparing the distribution attained from simulations to a normal distribution. Figure 6 indicates that the estimator appears to approximate a normal distribution for the Weibull and Gumbel domains of attraction, where the standard deviation reflects the theoretical value. In contrast, student- t distribution seems to be more dispersed, with a clear additional weight on the right hand side. The naïve estimator suffers the most from the kurtosis, while the average and weighted

average show less bias towards that direction and appear to conform closer to the normal approximation.

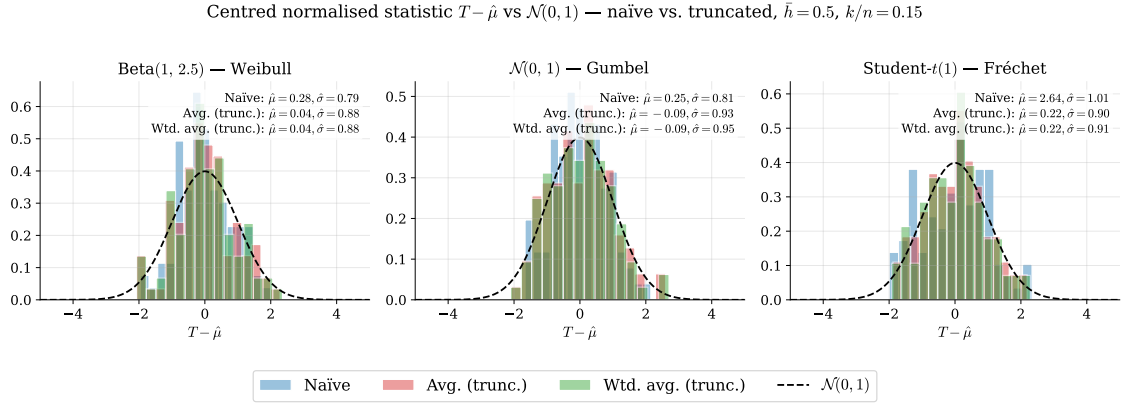


Figure 6: Histograms of the centred statistic $T - \hat{\mu}$ overlaid with $\mathcal{N}(0, 1)$, at $\bar{h} = 0.5$ and $k/n = 0.15$, for the naïve estimator and its two truncated competitors. The annotated $\hat{\mu}$ is the empirical mean of the uncentred T (an estimate of $-\lambda\bar{\Psi}_{\gamma,\rho}(1)$) and $\hat{\sigma}$ is its empirical standard deviation (theoretical target: 1).

We consider the QQ-plots comparing the simulated estimates to a normal distribution in Figure 7. The estimator follows the same pattern as in Figure 6, where the estimators derived from the Beta and normal distribution are relatively close to a Gaussian limit. The estimates from the normal distribution are more dispersed at the tails, as was reflected in the higher standard deviation that was observed before. There appears to be no clear difference among the three estimators for these two domains of attraction. The student- t distribution tells a rather different story, where there is a clear deviation from the normal distribution for the right-tail, as was observed before. Additionally, we see that the naïve estimator suffers in the left-tail as well, where both the average and weighted-average estimators perform well comparatively. This suggests that the inference based on a normal distribution is not a valid approach based on these estimates.

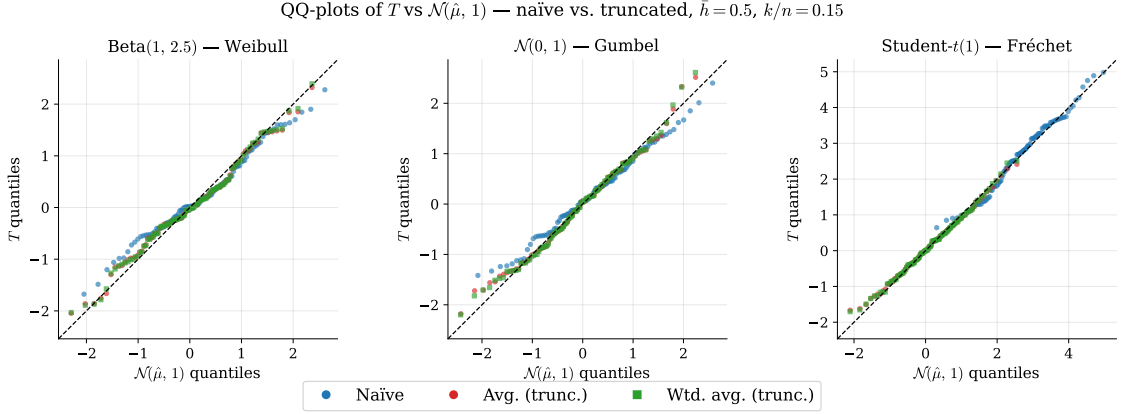


Figure 7: QQ-plots of T against $\mathcal{N}(\hat{\mu}, 1)$ quantiles at $\bar{h} = 0.5$ and $k/n = 0.15$. Fixing the variance at the theoretical value of one tests whether Lemma 3.2 is effective in practice; points on the diagonal confirm it.

4.2 Breaking Bandwidth

We have established that the naïve and truncated mean-based estimators seem to perform relatively well across the domains of attraction. We assume that the mean-based estimators are truncated relatively highly, corresponding to low order statistics being used exclusively. This section contrasts against previous results by comparing truncation against utilizing all order statistics, i.e. $\alpha = 0$. The simulations indicate that it still works for the Weibull and Gumbel domains of attraction, while the Fréchet domain of attraction becomes degenerate when the first moment does not exist.

We start by verifying the bandwidth condition for the untruncated bandwidth estimator. Figure 8 compares the estimated bandwidth against the oracle. The truncation does not seem to provide a qualitative change for the Weibull and Gumbel domains of attraction. The untruncated estimator has a slightly higher value, as the value converges to a finite expectation. The simulated pointwise confidence intervals remain relatively low, suggesting that most simulation fall inside of that range. This behaviour shifts as we consider the heavy-tailed student- t distribution, where the bandwidth values of the truncated and untruncated estimators differ by an order of magnitude. This is likely because the student- t distribution with 1 degrees of freedom, or Cauchy, does not admit a finite expectation if we do not truncate the integral. Hence, the value becomes rather large as it is, theoretically, diverging.

Bandwidth $h_n/a_0(n/k)$ vs power $\bar{h}$ — truncated vs. untruncated mean-based estimators, $k/n = 0.15$ (mean $\pm$ 80% interval)

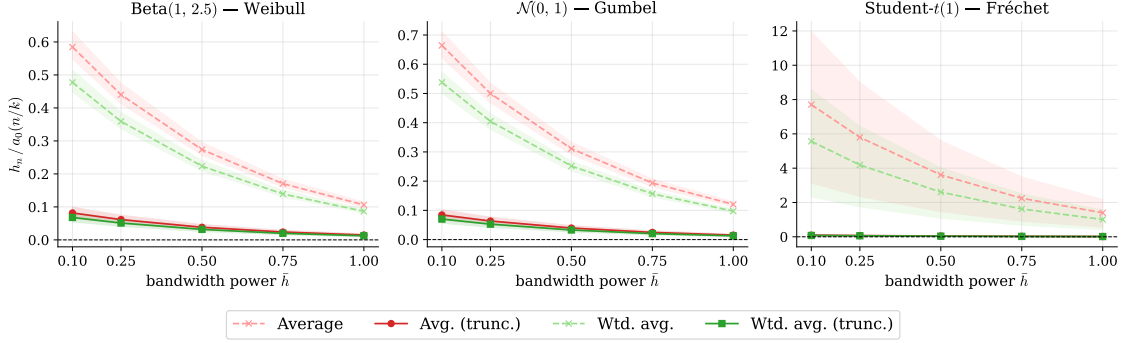


Figure 8: Ratio $h_n/a_0(n/k)$ as a function of bandwidth power $\bar{h}$ at $k/n = 0.15$, contrasting the untruncated average and weighted-average estimators (dashed) with their truncated counterparts (solid). Shading covers the 10th-90th percentile simulation interval.

Table 3 contains an overview of the results. For the Weibull and Gumbel domains of attraction, we see that the results for the truncated and untruncated estimators are hard to distinguish, providing similar bias, standard deviation, and RMSE. The Fréchet domain of attraction provides a clear difference, where the untruncated estimators have a larger bias standard deviation than their truncated counterparts. The degeneracy of the pareto distribution for an extreme value index of 1 likely explains this larger bias, as the bandwidth estimator is degenerate and the resulting loss function becomes a looser approximation of the pinball loss function. This means that it is more sensitive to large values than the traditional approach.

Distribution	Scale estimator	$\widehat{\mathbb{E}[T]}$	$\widehat{\text{Var}(T)}$	RMSE
Beta(1, 2.5)	Average	-0.001	0.870	0.866
	Avg. (trunc.)	-0.001	0.887	0.883
	Wtd. avg.	0.001	0.870	0.866
	Wtd. avg. (trunc.)	0.001	0.893	0.889
$\mathcal{N}(0, 1)$	Average	-0.144	0.902	0.909
	Avg. (trunc.)	-0.145	0.949	0.956
	Wtd. avg.	-0.144	0.905	0.912
	Wtd. avg. (trunc.)	-0.145	0.961	0.967
Student- $t(1)$	Average	11.843	4.701	12.734
	Avg. (trunc.)	0.610	1.426	1.545
	Wtd. avg.	9.670	5.428	11.076
	Wtd. avg. (trunc.)	0.399	1.241	1.297

Table 3: Empirical $\widehat{\mathbb{E}[T]}$, $\widehat{\text{Var}(T)}$, and $\text{RMSE} = \sqrt{\widehat{\mathbb{E}[T^2]}}$ at $\bar{h} = 0.5$, $k/n = 0.15$, $n = 2000$, 100 replications, contrasting untruncated and truncated average and weighted-average estimators.

We consider the direct consequence for the statistic T across bandwidth powers

in Figures 9 and 10. The results show a clear distinction between the domains of attraction. For the Weibull and Gumbel domains of attraction, we see that the three estimators have comparable bias, where the untruncated estimators have a slight reduced bias for higher powers of the bandwidths. Furthermore, they have a lower standard deviation across bandwidth powers, providing a clear advantage over the truncated estimators. However, this advantage vanishes for the student- t distribution, where the untruncated estimators have a bias and standard deviation that are both an order of magnitude higher. This holds uniformly across the bandwidth powers. This likely has to do with the degeneracy of the asymptotic untruncated expectation. The bandwidth explodes, by which the loss function becomes a poor approximation to the pinball loss function. Thus, large values would have a larger effect on the resulting estimator, which provides a rather vicious feedback cycle.

Empirical bias $\mathbb{E}[T]$ vs bandwidth power $\bar{h}$ — truncated vs. untruncated, $k/n=0.15$. Untruncated average/weighted-average estimators diverge for Student- t .

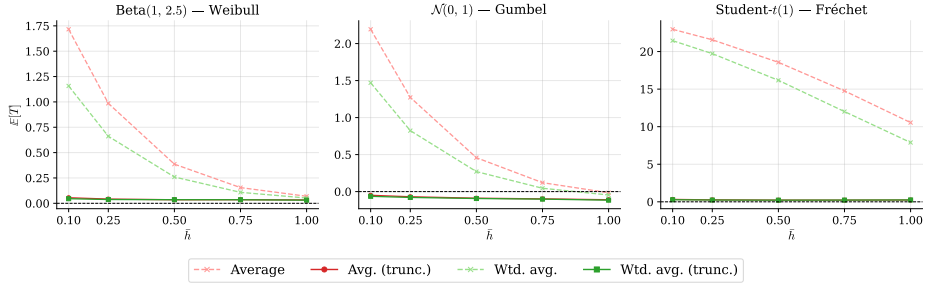


Figure 9: Empirical mean $\widehat{\mathbb{E}[T]}$ as a function of bandwidth power $\bar{h}$ at $k/n = 0.15$, contrasting untruncated (dashed) and truncated (solid) average and weighted-average estimators. The dashed horizontal line marks zero. Beta and Normal show no separation; the untruncated estimators diverge for Student- t .

Empirical SD(T) vs bandwidth power $\bar{h}$ — truncated vs. untruncated, $k/n = 0.15$ (reference line at 1)

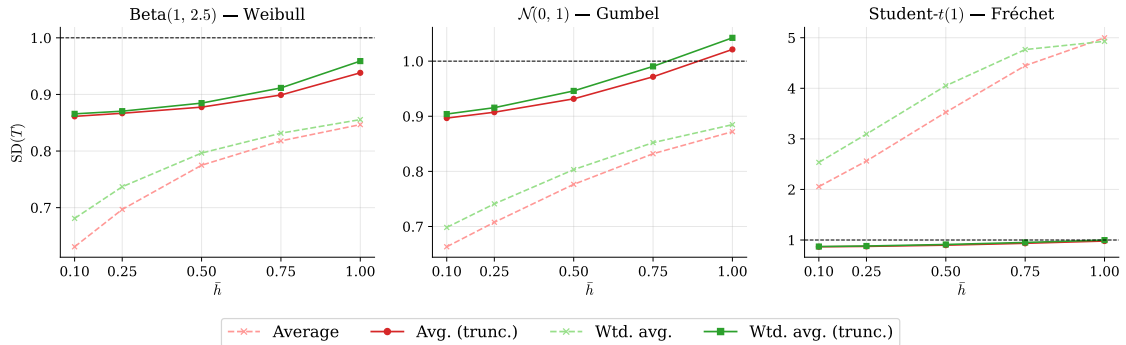


Figure 10: Empirical standard deviation $\widehat{\text{Var}(T)}$ as a function of bandwidth power $\bar{h}$ at $k/n = 0.15$. The dashed horizontal line marks the theoretical target of one. The untruncated estimators reach $\text{SD}(T) > 4$ for Student- t ; the truncated estimators stay below 1.5.

The bias and standard deviation mainly describe the concentration of the estimated parameters around the true parameters. However, for the purpose of inference, we need to consider the full distribution. Figure 11 shows the centred T -statistic at $\bar{h} = 0.5$ and $\frac{k}{n} = 0.15$. The results for the Beta and Gaussian distributions follow the standardized normal distribution closely, indicating that the asymptotic theory is valid. The untruncated and truncated estimators lie close together, indicating relatively little difference in their asymptotic performance. This behaviour breaks down for the student-t distribution, where the truncated estimators show a more heavy-tailed distribution than the expected standard normal. The untruncated estimators are far more dispersed, where their histograms do not clearly centre. If we consider their QQ-plots in Figure 12, this is even more clear. The untruncated estimators are far more heavy-tailed than the Gaussian distribution in both lower and upper tails. The results show that this is deviation of the Gaussian of an order of magnitude higher than the truncated estimators, indicating a breakdown of asymptotic theory. This likely remains an artifact of the diverging estimator as the associated expectation does not exist.

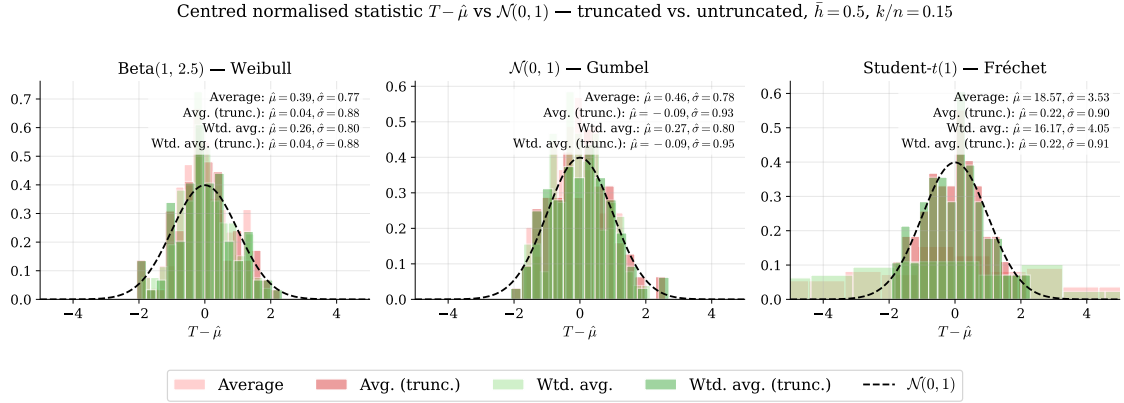


Figure 11: Histograms of the centred normalised statistic $T - \hat{\mu}$ overlaid with $\mathcal{N}(0, 1)$, at $\bar{h} = 0.5$ and $k/n = 0.15$, contrasting untruncated and truncated average and weighted-average estimators. The annotated $\hat{\mu}$ and $\hat{\sigma}$ are the empirical mean and standard deviation of the uncentred T for each estimator.

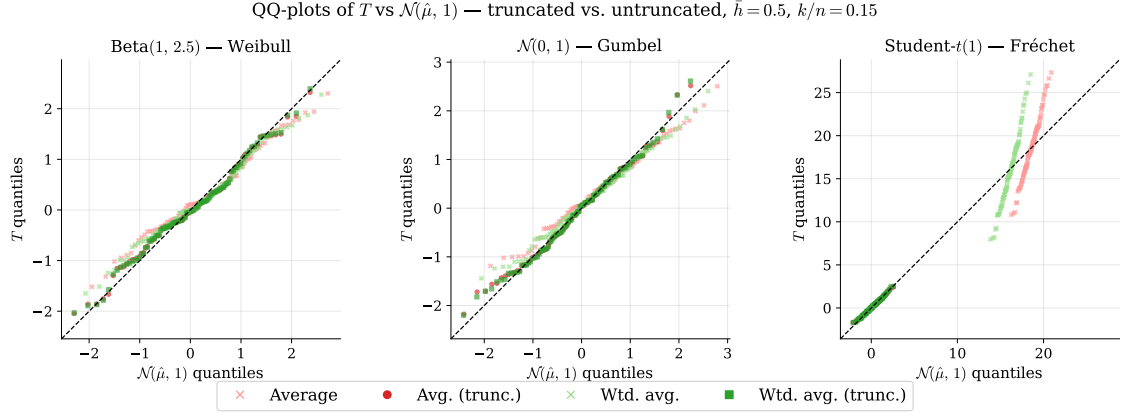


Figure 12: QQ-plots of T against $\mathcal{N}(\hat{\mu}, 1)$ quantiles at $\bar{h} = 0.5$ and $k/n = 0.15$, contrasting untruncated and truncated average and weighted-average estimators.

4.3 Asymptotic bias

The Burr12 distribution is a two-parameter heavy-tailed distribution where the extreme value index and the second order parameter can be derived explicitly in terms of the parameters of the distribution. This allows us to directly control the asymptotic bias that arises from the extreme value approximation. We run simulations for six combinations $\gamma_0 \in \{0.25, 0.50, 1.00\}$ and $\rho_0 \in \{-0.50, -1.00\}$ to investigate how the second order parameter governs the finite sample properties of the asymptotic bias $\lambda\bar{\Psi}_{\gamma,\rho}(1)$. We compare the naïve estimator against the two truncated mean-based estimators for $\alpha = 0.8$. The results are arranged in a 3×2 grid with rows indexed by γ_0 and columns by ρ_0 .

We verify the bandwidth condition by comparing the estimated bandwidths against the oracle bandwidth for the Burr12 distribution. Figure 13 contains the results, which show that the truncated estimators are lower than the naïve estimator, where all three seem to satisfy the bandwidth condition. This is consistent with the results for the heavy-tailed student-t distribution.

Burr12: bandwidth $h_n/a_0(n/k)$ vs power $\bar{h}$ — naïve vs. truncated, $k/n = 0.15$ (mean $\pm$ 80% interval)

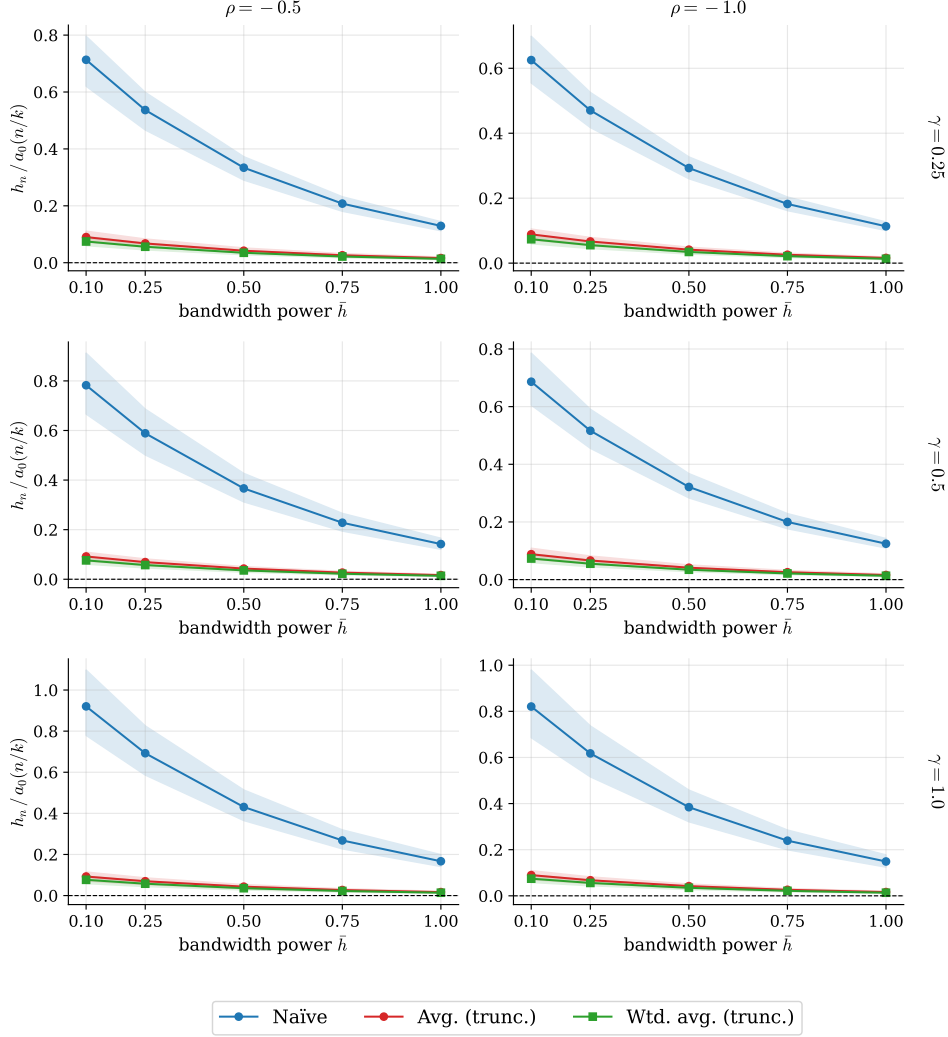


Figure 13: Ratio $h_n/a_0(n/k)$ for the six Burr12 configurations at $k/n = 0.15$, comparing the naïve estimator to its truncated average and weighted-average counterparts. Rows correspond to $\gamma_0 \in \{0.25, 0.50, 1.00\}$; columns to $\rho_0 \in \{-0.50, -1.00\}$. Shading covers the 10th–90th percentile simulation interval.

We consider the finite-sample concentration of the estimators around the true parameters. Table 4 provides a summary of the estimators. The results are not able to illuminate a clear pattern across the second order parameters, where the estimators or groups of estimators do not follow the same pattern. For the relatively light-tailed case of $\gamma_0 = 0.25$, we see that the mean-based estimators have a similar bias and variance, which are both lower than that of the naïve estimator. This contrasts against the $\gamma_0 = 0.5$ case, where the naïve estimator has a far lower bias than the mean-based estimator, leading to a lower RMSE overall. This difference flips for $\gamma_0 = 1$, where the naïve estimator has a high bias and a high standard deviation for $\rho_0 = -0.5$, while a stronger tail similarity leads to a lower bias and lower standard deviation. The mean-based estimators have a lower standard deviation.

tion for both cases, yet have a larger bias in the second case. This can be explained by the fact that the naïve estimator uses slightly higher order statistics than the mean-based estimators, thus using information that is farther in the tail.

γ_0	Estimator	$\rho_0 = -0.50$			$\rho_0 = -1.00$		
		$\widehat{\mathbb{E}}[T]$	$\widehat{\text{Var}}(T)$	RMSE	$\widehat{\mathbb{E}}[T]$	$\widehat{\text{Var}}(T)$	RMSE
0.25	Naïve	-0.369	1.465	1.503	-0.195	1.147	1.158
	Avg. (trunc.)	-0.187	1.044	1.055	-0.170	1.082	1.090
	Wtd. avg. (trunc.)	-0.048	1.099	1.094	-0.178	1.080	1.089
0.50	Naïve	-0.137	1.803	1.799	0.031	1.175	1.170
	Avg. (trunc.)	0.143	1.206	1.208	0.161	1.314	1.317
	Wtd. avg. (trunc.)	0.441	1.766	1.812	0.132	1.242	1.243
1.00	Naïve	1.698	3.141	3.557	-0.080	1.568	1.563
	Avg. (trunc.)	0.270	1.240	1.263	0.207	1.070	1.085
	Wtd. avg. (trunc.)	0.288	1.320	1.344	0.148	1.130	1.134

Table 4: Simulation summary for the six Burr12 configurations at reference settings $\bar{h} = 0.5$, $k/n = 0.15$, $n = 2000$, 100 replications, comparing the naïve estimator to its truncated average and weighted-average counterparts. Values are grouped by ρ_0 and ordered by γ_0 .

We expand on this overview by considering the bias and standard deviation more closely. Figures 14 and 15 show the bias and standard deviation of the estimators. The bias indicates that the naïve estimator often suffers from a large bias for low-bandwidths as opposed to the average and weighted average estimators, which are relatively close across the bandwidth values. There are two deviations from this trend, which both correspond to lighter tailed random variables. This can be explained by the lower second order parameter, which enforces a stronger self-similarity in the tail. This stronger self-similarity likely leads to less large deviations from the smooth tail approximation, thus providing a cleaner result.

Burr12: empirical bias $\mathbb{E}[T]$ vs bandwidth power $\bar{h}$ — naïve vs. truncated, $k/n = 0.15$

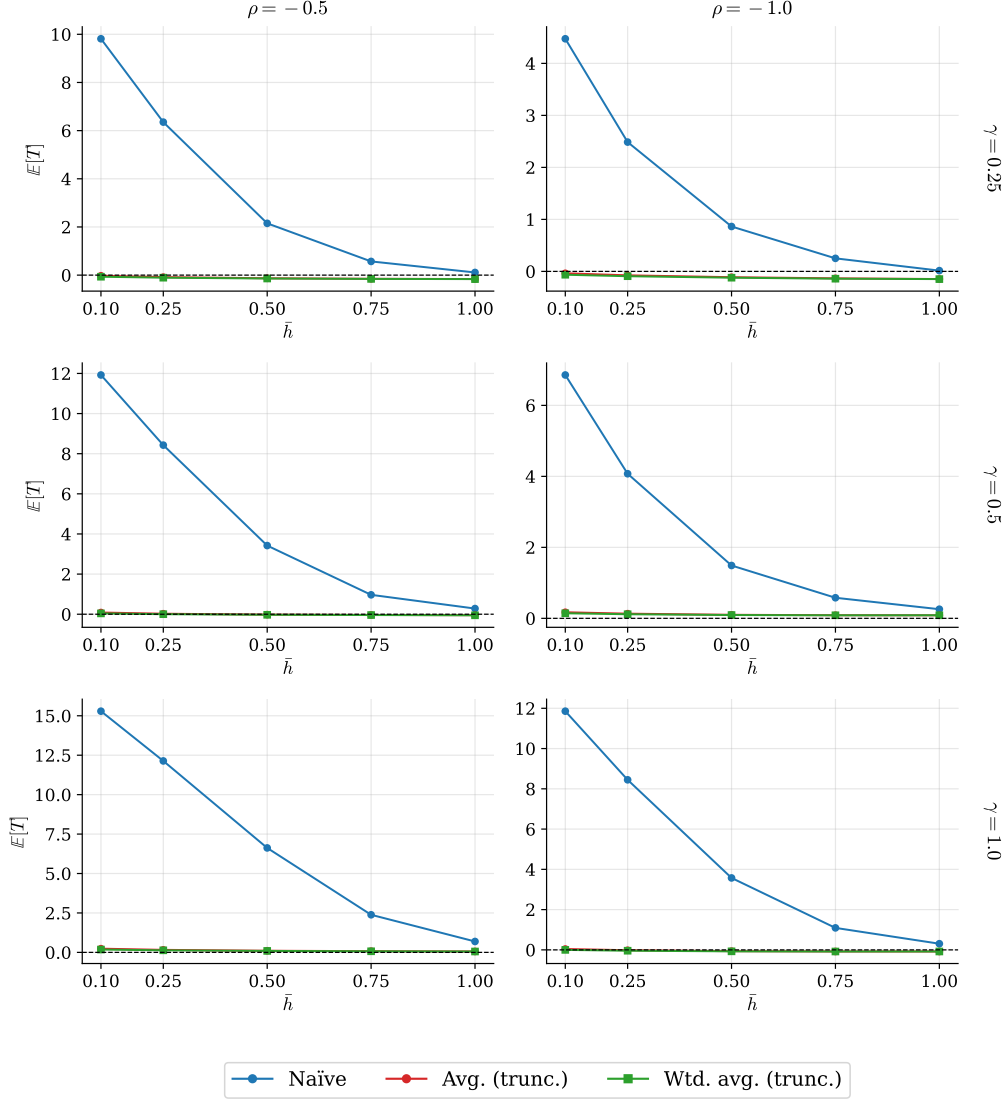


Figure 14: Empirical mean $\widehat{\mathbb{E}[T]}$ as a function of bandwidth power $\bar{h}$ for the six Burr12 configurations at $k/n = 0.15$, comparing the naïve estimator to its truncated average and weighted-average counterparts. The dashed line marks zero. Rows: $\gamma_0 \in \{0.25, 0.50, 1.00\}$; columns: $\rho_0 \in \{-0.50, -1.00\}$.

The standard deviation, results shown in Figure 15, follows a different pattern. We see that for the cases of $\gamma_0 \in \{0.25, 0.5\}$ there are no clear winners, where the mean-based estimators seem to perform better for low-bandwidth values, while the naïve estimator seems to work better for high-value bandwidths. This contrasts against the case of $\gamma_0 = 1$, the most heavy-tailed case, where the mean-based estimators clearly outperform the naïve estimator. Hence, the mean-based estimators trade-off performance for the light-tailed cases against robustness of the estimator in the heavy-tailed case. These patterns hold across the second order parameter. The standard deviation is likely not impacted by the strength of the self-similarity

as the quantile estimation performs well for continuous distributions, which is not dependent on the second order parameter.

Burr12: empirical $SD(T)$ vs bandwidth power $\bar{h}$ — naïve vs. truncated, $k/n=0.15$ (reference line at 1)

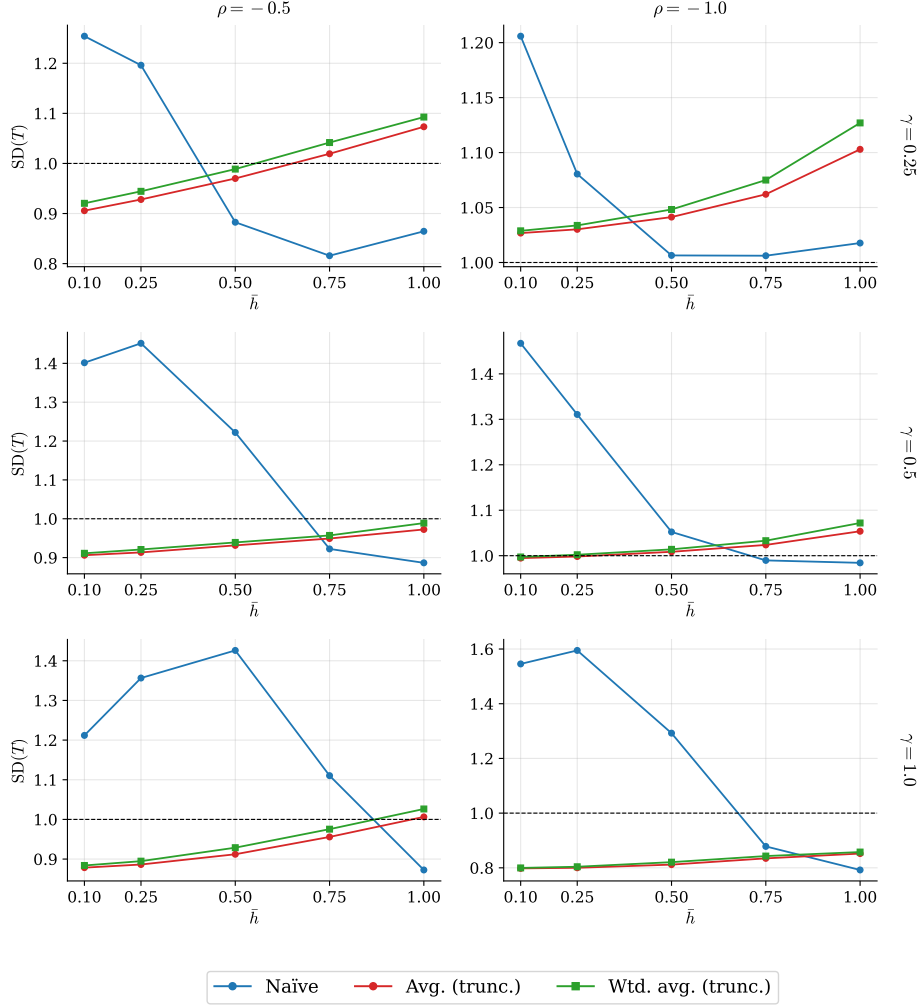


Figure 15: Empirical standard deviation $\widehat{\text{Var}}(T)$ as a function of bandwidth power $\bar{h}$ for the six Burr12 configurations at $k/n = 0.15$. The dashed line marks the theoretical target of one.

The concentration of the estimators around the true parameter mainly considers the consistency of the estimator. For the purpose of inference, we need to consider the wider distribution and the effect of the asymptotic bias. Figures 16 and 17 show the histograms and the QQ-plots comparing the T -statistic against the expected standard normal distribution. The results degenerate across the values of γ_0 , where the more heavy-tailed estimators deviate more from the reference Gaussian distribution. For the naïve estimator in particular, the QQ-plots show that for $\gamma_0 = 1$ the normal distribution is a poor descriptor for the distribution of the estimator. The mean-based estimators seem to improve upon this by a clear margin, where they mainly suffer from more heavy-tails, specifically a heavier upper tail. Furthermore,

we see that the estimators approximate the normal distribution more clearly for the lower second order parameters, suggesting that a strong tail self-similarity provides a cleaner estimation. This is likely because the stronger self-similarity enforces a stronger form of continuity, which eases the estimation process.

Burr12: centred $T - \hat{\mu}$ vs $\mathcal{N}(0, 1)$ — naïve vs. truncated, $\bar{h} = 0.5$, $k/n = 0.15$

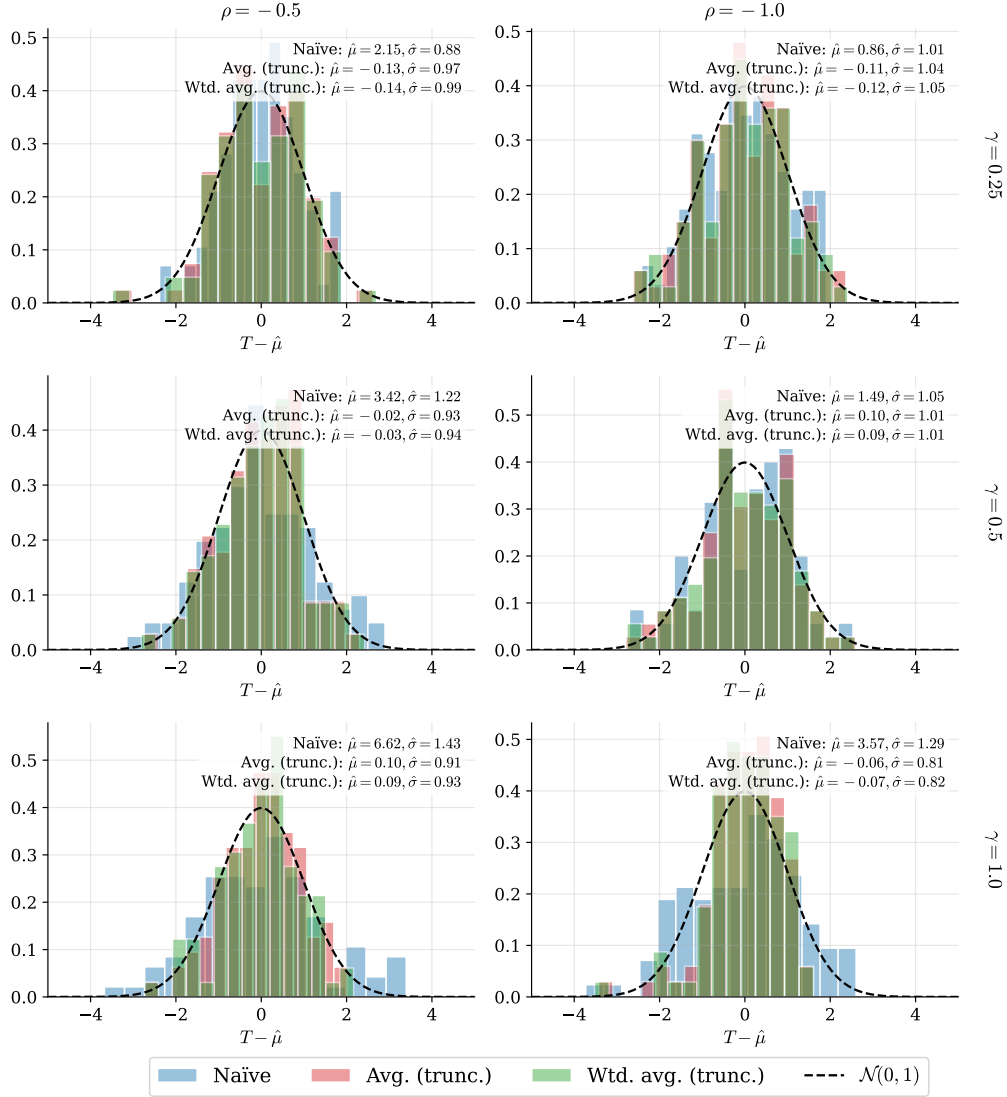


Figure 16: Histograms of the centred normalised statistic $T - \hat{\mu}$, overlaid with $\mathcal{N}(0, 1)$, comparing the naïve estimator to its truncated average and weighted-average counterparts. Settings: $\bar{h} = 0.5$, $k/n = 0.15$. Rows: $\gamma_0 \in \{0.25, 0.50, 1.00\}$; columns: $\rho_0 \in \{-0.50, -1.00\}$.

Burr12: QQ-plots of T vs $\mathcal{N}(\hat{\mu}, 1)$ — naïve vs. truncated, $\bar{h} = 0.5$, $k/n = 0.15$

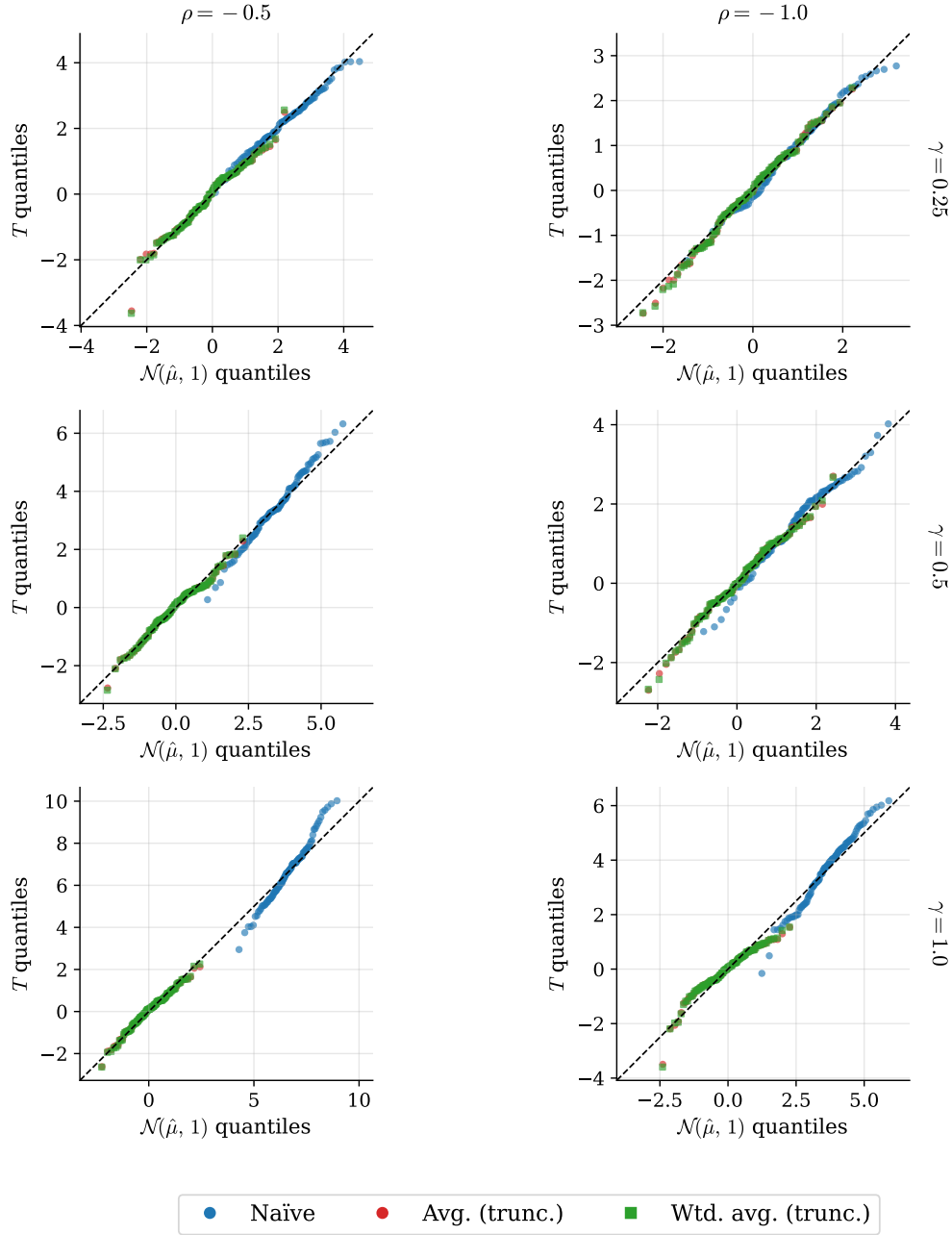


Figure 17: QQ-plots of T against $\mathcal{N}(\hat{\mu}, 1)$ quantiles at $\bar{h} = 0.5$ and $k/n = 0.15$ for the six Burr12 configurations.

5 Application

The simulation study of Section 4.1 onward validates the smoothed intermediate quantile estimator against known distributions. We apply the same estimator to the birthweight data introduced in the Section 1. We restrict attention to intermediate

levels $k/n \in [0.05, 0.25]$, which precludes extreme quantiles.

We use birthweight records from the 2024 U.S. natality survey (National Center for Health Statistics), where we investigate the birthweight field (grams). We filter out all "not-stated" sentinel code, which consists of 3,106 (0.09%) observations. We consider the left tail of the distribution, which corresponds to low birthweights. This is the most clinically relevant tail. We therefore flip the sign on birthweights, allowing us to directly make inference on the right tail. Every result can be translated back by flipping the sign again.

We utilize the same estimator configuration from Section 4.1. We omit the untruncated average and weighted-average estimators by their instability on Section 4.2's findings. We note that Figure 18 provides a positive indication that the bandwidth condition holds throughout. We note that the oracle estimator is unavailable. We substitute the naïve scale estimator, which is independent of the bandwidth power. The resulting ratio $h_n/\hat{f}_{\text{naive}}$ provides an approximation to the true ratio. Figure 18 shows the results for this ratio across the bandwidth power grid at each k/n . The ratio clearly stays below 1, where the truncated estimators are clearly lower than the naïve one across the spectrum. The difference is of the same order of magnitude as observed in the simulation study.

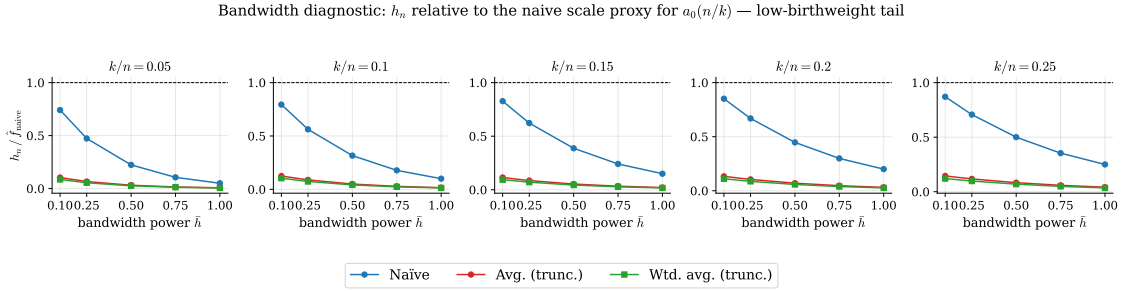


Figure 18: Ratio $h_n/\hat{f}_{\text{naive}}$ as a function of bandwidth power $\bar{h}$, for each tail fraction k/n , low-birthweight tail of the 2024 U.S. natality data. A ratio below one indicates the bandwidth condition of Theorem 3.3 is satisfied; $\hat{f}_{\text{naive}}$ is a data-driven proxy for the unknown oracle $a_0(n/k)$.

The results are shown in Table 5. The results indicate that the estimators are rather close across bandwidth estimators. The three bandwidths show a clear difference, suggesting that the interpolation among the observed quantiles provides a different approximation than the order statistic. This provides a principled approach to interpolate among order statistics within applications.

Scale estimator	Estimate (g)	90% CI low (g)	90% CI high (g)	Raw order stat. (g)
Naïve	2717.8	2716.9	2718.7	2730.0
Avg. (trunc.)	2733.5	2732.6	2734.4	2730.0
Wtd. avg. (trunc.)	2733.6	2732.7	2734.5	2730.0

Table 5: Estimated low-birthweight quantile at $\bar{h} = 0.5$, $k/n = 0.15$ ($n = 3,635,330$), with asymptotic 90% confidence interval, against the raw order statistic.

We consider the stability of the estimator across bandwidth values. The results are shown in Figure 19. The results show that the naïve estimator changes the most across bandwidth values, while the truncated estimators remain stable. This reflects the pattern we saw in the simulations, validating our approach empirically. The estimators converge to the order statistic for smaller and smaller bandwidths, as expected theoretically.

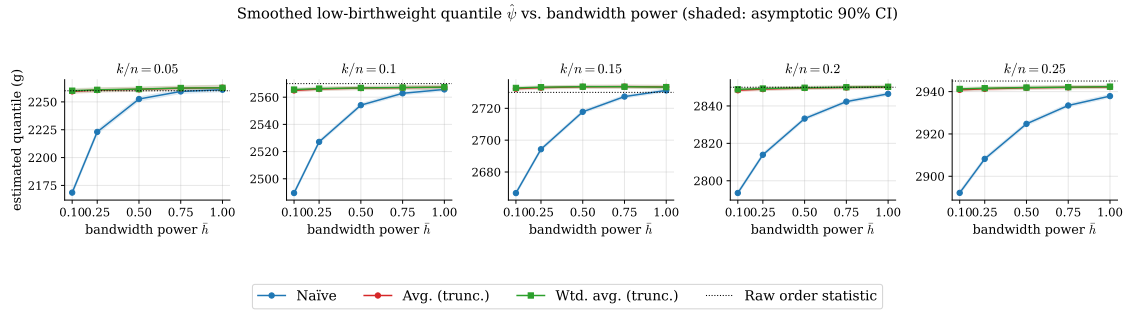


Figure 19: Estimated low-birthweight quantile $\hat{\psi}$ (grams) as a function of bandwidth power $\bar{h}$, for each tail fraction k/n . Shading shows the asymptotic 90% confidence interval obtained by inverting $T \rightarrow N(0, 1)$ (the bias term is dropped, as it is unidentifiable without a known data-generating process); the dotted line marks the raw, unsmoothed order statistic.

We compare the estimators across tail fractions in Figure 20 at a reference bandwidth of $\bar{h} = 0.5$. The truncated estimator are very close to the raw-order statistic at each level, while the naïve estimator stays below both consistency across levels. The gap widens at higher quantile levels, likely because the naïve estimator utilizes observations deeper in the tail to construct the bandwidth.

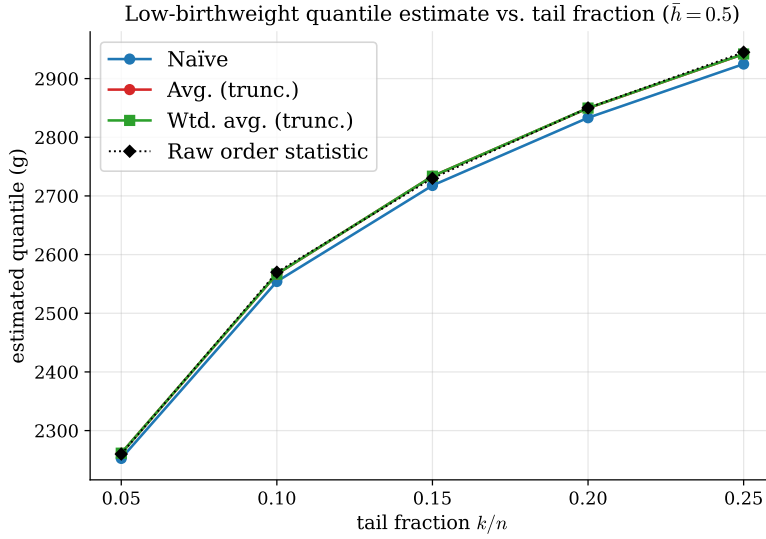


Figure 20: Estimated low-birthweight quantile (grams) against tail fraction k/n at $\bar{h} = 0.5$, comparing the naïve, truncated average, and truncated weighted-average estimators against the raw order statistic.

The empirical results follow the same pattern as in the simulation study. The truncated mean-based estimators are more robust to the bandwidth choice than the naïve estimator, where both remain close to the order statistic. This provides a clear motivation and validation of applying the methodology to real-world data. However, this is not without limitation, as is discussed in Section 6.

6 Conclusion

We investigate the generalization of the smoothed quantile estimation through convolution smoothing to intermediate quantile estimation. We investigate this methodology under kernels bounded by sub-Gaussian tails, generalizing the class of kernels originally used in Fernandes, Guerre, and Horta (2021). We show that local consistency and local asymptotic normality for the resulting estimator hold under different bandwidth regimes. The results show, explicitly, that the bandwidth has to be related to the rate of convergence of large values towards the tail shape. We additionally show that, under appropriate rates, the classical kernel density estimator centered at a consistent estimator of the quantile provides a consistent estimator of the rate function. This provides an unbiased estimator of the theoretical rate function, which, to the author’s knowledge, has not yet been established in the literature.

We verify our approach through a Monte Carlo study. The results indicate that the methodology works well for finite and light-tailed distributions. This is true for both the smoothed quantile estimation and the normalization by the second order derivative. The heavy-tailed distributions turn out to be the problematic case. The smoothed quantile estimation seems to work well. However, the results are skewed by the normalization being inadequate, as large deviations

from the estimated or true quantile ensure that relatively little weight actually centres on these quantiles, which themselves are low probability events. This ensures that the normalization blows up, thus providing worse results. This highlights a practical limitation of the methodology, which has to be addressed by an appropriate estimator of the bandwidth and the associated kernel.

The methodology has a few different limitations that can be addressed in future work. First, we assume independent and identically distributed data for quantile estimation. The theory has classically been extended to settings with dependence, see Chernozhukov, Fernández-Val, and Kaji (2016), which often occur within economic settings. Hence, expanding the tail empirical process theory that belies the estimator to allow for dependence would provide a clear research direction. Second, we assume that there are no external factors influencing the extremes. This differs from allowing for covariate dependence, such as in Chernozhukov, Fernández-Val, and Kaji (2017), through quantile regression. The extension to such a dependence would provide a novel method to perform quantile regression and more general quantile inference in tail regions. This directions can be approached by expanding the tail empirical process work to include weighted sums of functions indexed by covariates. This is clearly reflected in the application, where birthweights are affected by a number of different factors that are not taken into account at this point. Third, we unable to quantify the effect of contamination within our estimation. The topic of contamination of data within such a setting is rather delicate, as we need to be able to distinguish between an "outlier" and an "extreme event". A candidate for such research is the work of Ke and Medina (2026), which provides a general measure for the effect of contamination on an estimator. For our case, this formulation is particularly well suited as the breakdown-point can be adjusted. By considering the similarity between such a breakdown point and the bandwidth we work with, we conjecture that the breakdown point for extremes would shrink at rate $a_0(n/k)$ for classical quantile regression, and faster for our kernel smoothed approach.

The main limitation of the results is the normalization following from the bandwidth and kernel choices. This highlights the importance of developing an appropriate set of bandwidth estimator, specifically tailored for extreme quantiles. The naïve estimator still works relatively well, but it does not utilize all available information, as it is only characterized by two order statistics. The mean-based estimators for the bandwidth seem to break down, likely due to the non-existence of the associated moment. This is partially addressed through the truncating of the mean-based estimators. However, further research is possible into more natural bandwidth choices.

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Appendix

This appendix contains the supporting theoretical results of Section 3 and the generative AI statement.

A Proofs

Lemma A.1. *Suppose that the distribution satisfies the second order condition given in (6). Assume Condition 4 holds and that $h_n^{-1} = o(|\log(k/n)|^{-1/2})$ as $n \rightarrow \infty$. Then, we have that for $r < \frac{1}{|\gamma_0|}$ that*

$$\sup_{\phi \in B_r(0)} \left| \int_{-\infty}^{\infty} \frac{n}{k} (1 - \mathbb{F}_n) \left(a_0 \left(\frac{n}{k} \right) (\phi - z) + U \left(\frac{n}{k} \right) \right) K_{h_n}(z) dz - D_{\gamma_0}(\phi) \right| \xrightarrow{P} 0. \quad (9)$$

Proof. Consider $r, c > 0$ such that $r + c < \frac{1}{|\gamma_0|}$. Then, we have that

$$\begin{aligned} & \int_{-\infty}^{\infty} \frac{n}{k} (1 - \mathbb{F}_n) \left(a_0 \left(\frac{n}{k} \right) (\phi - z) + U \left(\frac{n}{k} \right) \right) K_{h_n}(z) dz \\ &= \int_{-\infty}^{\infty} \mathbb{1}\{|z| \leq c\} \frac{n}{k} (1 - \mathbb{F}_n) \left(a_0 \left(\frac{n}{k} \right) (\phi - z) + U \left(\frac{n}{k} \right) \right) K_{h_n}(z) dz \\ & \quad + \int_{-\infty}^{\infty} \mathbb{1}\{|z| > c\} \frac{n}{k} (1 - \mathbb{F}_n) \left(a_0 \left(\frac{n}{k} \right) (\phi - z) + U \left(\frac{n}{k} \right) \right) K_{h_n}(z) dz \\ &= J_1 + J_2. \end{aligned}$$

We first consider J_2 . We consider the absolute value, by which we have that

$$\begin{aligned} |J_2| &\leq \int_{\{|zh_n| > c\}} \frac{n}{k} (1 - \mathbb{F}_n) \left(a_0 \left(\frac{n}{k} \right) (\phi - z) + U \left(\frac{n}{k} \right) \right) |K(z)| dz \\ &\leq \frac{n}{k} \int_{\{|zh_n| > c\}} |K(z)| dz. \end{aligned}$$

Then, under the assumption that $h_n \rightarrow 0$, we can find n_0 such that for all $n \geq n_0$ by Condition 4, we have that

$$\begin{aligned} |J_2| &\leq \frac{n}{k} \int_{\{|zh_n| > c\}} |K(z)| dz \leq C_K \frac{n}{k} \int_{\{|zh_n| > c\}} \exp(-c_K z^2) dz \\ &\leq C_K \exp \left(- \left[c_K \frac{c^2}{h_n^2} - \log(n/k) \right] \right), \end{aligned}$$

where the last inequality is derived from Mill's ratio. Therefore, by the assumption on h , this term is controlled. This deals with the second term, by which the first term remains. We have

$$\begin{aligned} J_1 &= \int_{-\infty}^{\infty} \mathbb{1}\{|z| \leq c\} \frac{n}{k} (1 - \mathbb{F}_n) \left(a_0 \left(\frac{n}{k} \right) (\phi - z) + U \left(\frac{n}{k} \right) \right) K_{h_n}(z) dz \\ &= \int_{-\infty}^{\infty} \mathbb{1}\{|z| \leq c\} D_{\gamma_0}(\phi - z) K_{h_n}(z) dz + O_P \left(\frac{1}{\sqrt{k}} \right) \int_{-\infty}^{\infty} \mathbb{1}\{|z| \leq c\} K_{h_n}(z) dz \\ &= \int_{-\infty}^{\infty} \mathbb{1}\{|zh_n| \leq c\} [D_{\gamma_0}(\phi - zh_n) - D_{\gamma_0}(\phi)] K(z) dz \\ & \quad + \left[O_P \left(\frac{1}{\sqrt{k}} \right) + D_{\gamma_0}(\phi) \right] \int_{-\infty}^{\infty} \mathbb{1}\{|z| \leq c\} K_{h_n}(z) dz. \end{aligned}$$

Then, by the mean-value theorem, there exists $\tau \in (0, 1)$ such that

$$\begin{aligned} J_1 &= h_n \int_{-\infty}^{\infty} \mathbb{1}\{|zh_n| \leq c\} D'_{\gamma_0}(\phi - zh_n\tau) K(z) dz \\ &\quad + \left[O_P\left(\frac{1}{\sqrt{k}}\right) + D_{\gamma_0}(\phi) \right] \int_{-\infty}^{\infty} \mathbb{1}\{|z| \leq c\} K_{h_n}(z) dz. \end{aligned}$$

Then, as $r + c < \frac{1}{|\gamma_0|}$ and $h_n \rightarrow 0$ as $n \rightarrow \infty$, we see that the above converges to $D_{\gamma_0}(\phi)$. $\square$

Lemma A.2. *Suppose that the quantile function of X satisfies the second order condition given in (6). Further, under Condition 4, assume that $h_n^{-1} = o\left(|\log(\sqrt{k}/n)|^{-1/2}\right)$ and $h_n^{\ell+1}\sqrt{k} = o(1)$ as $n \rightarrow \infty$. Then, we have that*

$$\sup_{\phi \in B_r(0)} \left| \sqrt{k} \left(\int_{-\infty}^{\infty} \left[\frac{n}{k} (1 - \mathbb{F}_n) \left(a_0 \left(\frac{n}{k} \right) (\phi - z) + U \left(\frac{n}{k} \right) \right) - D_{\gamma_0}(\phi) \right] K_{h_n}(z) dz \right) \right| \quad (10)$$

$$- W_n(D_{\gamma_0}(\phi)) - \sqrt{k} A_0 \left(\frac{n}{k} \right) (D_{\gamma_0}(\phi))^{1+\gamma_0} \overline{\Psi}_{\gamma_0, \rho_0} \left(\frac{1}{D_{\gamma_0}(\phi)} \right) \Big| = o_P(1). \quad (11)$$

Proof. Theorem 2.1 provides the main work which we consider here. Thus, we are mainly interested in showing that

$$\sqrt{k} \left(\int_{-\infty}^{\infty} \left[\frac{n}{k} (1 - \mathbb{F}_n) \left(a_0 \left(\frac{n}{k} \right) (\phi - z) + U \left(\frac{n}{k} \right) \right) - D_{\gamma_0}(\phi) \right] K_{h_n}(z) dz \right)$$

has the same limit as the original process in Theorem 2.1. We can write that

$$\begin{aligned} & \int_{-\infty}^{\infty} \sqrt{k} \left[\frac{n}{k} (1 - \mathbb{F}_n) \left(a_0 \left(\frac{n}{k} \right) (\phi - z) + U \left(\frac{n}{k} \right) \right) - D_{\gamma_0}(\phi) \right] K_{h_n}(z) dz \\ &= \int_{-\infty}^{\infty} \sqrt{k} \mathbb{1}\{|z| \leq c\} \left[\frac{n}{k} (1 - \mathbb{F}_n) \left(a_0 \left(\frac{n}{k} \right) (\phi - z) + U \left(\frac{n}{k} \right) \right) - D_{\gamma_0}(\phi) \right] K_{h_n}(z) dz \\ &+ \int_{-\infty}^{\infty} \sqrt{k} \mathbb{1}\{|z| > c\} \left[\frac{n}{k} (1 - \mathbb{F}_n) \left(a_0 \left(\frac{n}{k} \right) (\phi - z) + U \left(\frac{n}{k} \right) \right) - D_{\gamma_0}(\phi) \right] K_{h_n}(z) dz \\ &= J_1 + J_2. \end{aligned}$$

We first consider J_2 . We consider the absolute value, where we can write that

$$\begin{aligned} |J_2| &\leq \int_{-\infty}^{\infty} \sqrt{k} \mathbb{1}\{|z| > c\} \left| \frac{n}{k} (1 - \mathbb{F}_n) \left(a_0 \left(\frac{n}{k} \right) (\phi - z) + U \left(\frac{n}{k} \right) \right) - D_{\gamma_0}(\phi) \right| |K_{h_n}(z)| dz \\ &\leq \left(1 + \frac{k}{n} D_{\gamma_0}(\phi) \right) \sqrt{k} \frac{n}{k} \int_{\{|zh_n| > c\}} |K(z)| dz \\ &\leq C_K \left(1 + \frac{k}{n} D_{\gamma_0}(\phi) \right) \sqrt{k} \frac{n}{k} \int_{\{|zh_n| > c\}} \exp(-c_K(z^2)) dz \\ &\leq C_K \left(1 + \frac{k}{n} D_{\gamma_0}(\phi) \right) \exp \left(- \left[c_K \left(\frac{c}{h_n} \right)^2 - \log \left(\frac{n}{k} \right) - \frac{1}{2} \log(k) \right] \right), \end{aligned}$$

where the last inequality arises by Mill's ratio inequality. Therefore, we can control the first term by the assumption. Hence, we only need to control the first term. We can write that

$$\begin{aligned}
J_1 &= \int_{-\infty}^{\infty} \sqrt{k} \mathbb{1}\{|z| \leq c\} \left[\frac{n}{k} (1 - \mathbb{F}_n) \left(a_0 \left(\frac{n}{k} \right) (\phi - z) + U \left(\frac{n}{k} \right) \right) - D_{\gamma_0}(\phi) \right] K_{h_n}(z) dz \\
&= \int_{-\infty}^{\infty} \sqrt{k} \mathbb{1}\{|z| \leq c\} \left[\frac{n}{k} (1 - \mathbb{F}_n) \left(a_0 \left(\frac{n}{k} \right) (\phi - z) + U \left(\frac{n}{k} \right) \right) - D_{\gamma_0}(\phi - z) \right] K_{h_n}(z) dz \\
&\quad + \int_{-\infty}^{\infty} \sqrt{k} \mathbb{1}\{|z| \leq c\} [D_{\gamma_0}(\phi - z) - D_{\gamma_0}(\phi)] K_{h_n}(z) dz \\
&= J_{1.1} + J_{1.2}.
\end{aligned}$$

For $J_{1.1}$, we can utilize Theorem 2.1 to be able to write

$$\begin{aligned}
&\int_{-\infty}^{\infty} \sqrt{k} \mathbb{1}\{|z| \leq c\} \left[\frac{n}{k} (1 - \mathbb{F}_n) \left(a_0 \left(\frac{n}{k} \right) (\phi - z) + U \left(\frac{n}{k} \right) \right) - D_{\gamma_0}(\phi - z) \right] K_{h_n}(z) dz \\
&= \int_{-\infty}^{\infty} \mathbb{1}\{|z| \leq c\} [W_n(D_{\gamma_0}(\phi - z)) + D_{\gamma_0}(\phi - z)^{1+\gamma_0} \bar{\Psi}_{\gamma_0, \rho_0}(1/D_{\gamma_0}(\phi - z))] K_{h_n}(z) dz \\
&\quad + o_P(1) \int_{-\infty}^{\infty} \mathbb{1}\{|z| \leq c\} K_{h_n}(z) dz.
\end{aligned}$$

Then, these are asymptotically equivalent to their non-smoothed counterparts by the modulus of continuity of the Brownian motion and the smoothness of the function $x \mapsto D_{\gamma_0}(x)^{1+\gamma_0} \bar{\Psi}_{\gamma_0, \rho_0}(1/D_{\gamma_0}(x))$. Hence, we only need to control the term $J_{1.2}$. To this end, we consider the ℓ -th order kernel assumption from Condition 1.B. We have

$$\begin{aligned}
J_{1.2} &= \int_{-\infty}^{\infty} \sqrt{k} \mathbb{1}\{|z| \leq c\} [D_{\gamma_0}(\phi - z) - D_{\gamma_0}(\phi)] K_{h_n}(z) dz \\
&= \int_{-\infty}^{\infty} \sqrt{k} \mathbb{1}\{|zh_n| \leq c\} \left[\sum_{l=1}^{\ell} D_{\gamma_0}^{(l)}(\phi) \frac{(zh_n)^l}{l!} \right] K(z) dz \\
&\quad + \int_{-\infty}^{\infty} \sqrt{k} \mathbb{1}\{|zh_n| \leq c\} \left[D_{\gamma_0}^{(\ell+1)}(\phi + \nu zh_n) \frac{(zh_n)^{\ell+1}}{(\ell+1)!} \right] K(z) dz.
\end{aligned}$$

Then, utilizing the fact that K is an ℓ -th order kernel, we can rewrite the first term as

$$\int_{-\infty}^{\infty} \sqrt{k} \mathbb{1}\{|zh_n| \leq c\} \left[D_{\gamma_0}^{(l)}(\phi) \frac{(zh_n)^l}{l!} \right] K(z) dz = \frac{D_{\gamma_0}^{(l)}(\phi)}{l!} h_n^l \sqrt{k} \int_{-\infty}^{\infty} \mathbb{1}\{|zh_n| > c\} z^l K(z) dz.$$

Then, for $l = 1, \dots, \ell$, as $D_{\gamma_0}^{(l)}(\phi)$ is well defined as $|\phi| \leq r < \frac{1}{|\gamma_0|}$, we require that

$$h_n^l \sqrt{k} \int_{-\infty}^{\infty} \mathbb{1}\{|zh_n| > c\} z^l K(z) dz \rightarrow 0, \quad h_n^{\ell+1} \sqrt{k} \rightarrow 0,$$

as $n \rightarrow \infty$. We note that

$$\begin{aligned} \left| h_n^l \sqrt{k} \int_{-\infty}^{\infty} \mathbb{1}\{|zh_n| > c\} z^l K(z) dz \right| &\leq h_n^l \sqrt{k} \int_{-\infty}^{\infty} \mathbb{1}\{|zh_n| > c\} \frac{c^{\ell+1-l}}{c^{\ell+1-l}} |z^l K(z)| dz \\ &\leq h_n^l \sqrt{k} h_n^{\ell+1-l} \int_{-\infty}^{\infty} \left| \frac{z^{\ell+1-l}}{c^{\ell+1-l}} z^l K(z) \right| dz = h_n^{\ell+1} \sqrt{k} \int_{\{|zh_n| > c\}} \left| \frac{1}{c^{\ell+1-l}} z^{\ell+1} K(z) \right| dz. \end{aligned}$$

Then, as $c > 0$ and we have an ℓ -th order kernel, this convergence is implied by $h_n^{\ell+1} \sqrt{k} \rightarrow 0$ as $n \rightarrow \infty$. $\square$

Lemma A.3. *Suppose that the quantile function of X satisfies the second order condition given in (6). Assume Condition 4 holds and that $h_n^{-1} = o\left(|\log(\sqrt{k}/n)|^{-1/2}\right)$ as $n \rightarrow \infty$. Then, we have that for $\phi \in \overline{B}_r(0)$ with $r < \frac{1}{|\gamma_0|}$*

$$\frac{1}{n} \sum_{i=1}^n \frac{n}{k} K_{h_n} \left(\phi - \frac{Z_i - U(n/k)}{a_0(n/k)} \right) \xrightarrow{P} -D'_{\gamma_0}(\phi),$$

as $n \rightarrow \infty$.

Proof. To start, we note that the continuous nature of K means that we can express this function in terms of the tail empirical process through the fundamental lemma of calculus, as

$$\begin{aligned} &\frac{1}{n} \sum_{i=1}^n \frac{n}{k} K_{h_n} \left(\phi - \frac{Z_i - U(n/k)}{a_0(n/k)} \right) \\ &= \frac{1}{n} \sum_{i=1}^n \frac{n}{k} \int_{-\infty}^{\infty} \mathbb{1}\left\{Z_i > za_0\left(\frac{n}{k}\right) + U\left(\frac{n}{k}\right)\right\} \frac{1}{h_n^2} K' \left(\frac{\phi - z}{h_n} \right) dz \\ &= \frac{1}{n} \sum_{i=1}^n \frac{n}{k} \int_{-\infty}^{\infty} \mathbb{1}\left\{Z_i > (\phi - zh_n)a_0\left(\frac{n}{k}\right) + U\left(\frac{n}{k}\right)\right\} \frac{1}{h_n} K'(z) dz \\ &= \frac{1}{n} \sum_{i=1}^n \frac{n}{k} \int_{-\infty}^{\infty} \mathbb{1}\{|zh_n| \leq c\} \mathbb{1}\left\{Z_i > (\phi - zh_n)a_0\left(\frac{n}{k}\right) + U\left(\frac{n}{k}\right)\right\} \frac{1}{h_n} K'(z) dz \\ &\quad + \frac{1}{n} \sum_{i=1}^n \frac{n}{k} \int_{-\infty}^{\infty} \mathbb{1}\{|zh_n| > c\} \mathbb{1}\left\{Z_i > (\phi - zh_n)a_0\left(\frac{n}{k}\right) + U\left(\frac{n}{k}\right)\right\} \frac{1}{h_n} K'(z) dz \\ &= J_1 + J_2, \end{aligned}$$

where $c > 0$ is a constant such that $r + c < \frac{1}{|\gamma_0|}$. Then, we can utilize Theorem 2.1 by the fact that $\phi - zh_n$ lies in the set of admissible values, such that

$$\begin{aligned} J_1 &= \frac{1}{n} \sum_{i=1}^n \frac{n}{k} \int_{-\infty}^{\infty} \mathbb{1}\{|zh_n| \leq c\} \mathbb{1}\left\{Z_i > (\phi - zh_n)a_0\left(\frac{n}{k}\right) + U\left(\frac{n}{k}\right)\right\} \frac{1}{h_n} K'(z) dz \\ &= o_P\left(\frac{1}{\sqrt{k}}\right) \int_{-\infty}^{\infty} \mathbb{1}\{|zh_n| \leq c\} \frac{1}{h_n} K'(z) dz + \int_{-\infty}^{\infty} \mathbb{1}\{|zh_n| \leq c\} D_{\gamma_0}(\phi - zh_n) \frac{1}{h_n} K'(z) dz \\ &= J \end{aligned}$$

Then, using a change of variables and partial integration, we can rewrite the second term as follows:

$$\begin{aligned}
J &= o_P(1) \frac{K\left(\frac{c}{h_n}\right) - K\left(-\frac{c}{h_n}\right)}{h_n \sqrt{k}} + \int_{-c}^c D_{\gamma_0}(\phi - z) \frac{1}{h_n^2} K'\left(\frac{z}{h_n}\right) dz \\
&= o_P(1) \frac{K\left(\frac{c}{h_n}\right) - K\left(-\frac{c}{h_n}\right)}{h_n \sqrt{k}} + \left[D_{\gamma_0}(\phi - z) \frac{1}{h_n} K\left(\frac{z}{h_n}\right) \right]_{-c}^c \\
&\quad - \int_{-c}^c D'_{\gamma_0}(\phi - z) \frac{1}{h_n} K\left(\frac{z}{h_n}\right) dz \\
&= o_P(1) \frac{K\left(\frac{c}{h_n}\right) - K\left(-\frac{c}{h_n}\right)}{h_n \sqrt{k}} + D_{\gamma_0}(\phi - c) \frac{K\left(\frac{c}{h_n}\right)}{h_n} - D_{\gamma_0}(\phi + c) \frac{K\left(-\frac{c}{h_n}\right)}{h_n} \\
&\quad - \int_{-\frac{c}{h_n}}^{\frac{c}{h_n}} [D'_{\gamma_0}(\phi - zh_n) - D'_{\gamma_0}(\phi)] K(z) dz - D'_{\gamma_0}(\phi) \int_{-\frac{c}{h_n}}^{\frac{c}{h_n}} K(z) dz.
\end{aligned}$$

Then, we note that the first three terms converge to zero by the assumption on h and K . Note that as $\phi \pm c \in B_{r+c}(0)$ with $r + c < \frac{1}{|\gamma_0|}$, that the function $D_{\gamma_0}(\phi \pm c)$ is finite. Thus, we consider the latter two terms. First, note that as $h_n \rightarrow 0$ $n \rightarrow \infty$ and the fact that K is a kernel function that $\int_{-\frac{c}{h_n}}^{\frac{c}{h_n}} K(z) dz \rightarrow 1$ as $n \rightarrow \infty$. Then, we can write that

$$\begin{aligned}
&\int_{-\frac{c}{h_n}}^{\frac{c}{h_n}} [D'_{\gamma_0}(\phi - zh_n) - D'_{\gamma_0}(\phi)] K(z) dz \\
&= \sum_{j=1}^{\ell} D_{\gamma_0}^{(j)}(\phi) h_n^j \int_{-\frac{c}{h_n}}^{\frac{c}{h_n}} z^j K(z) dz + h_n^{\ell+1} \int_{-\frac{c}{h_n}}^{\frac{c}{h_n}} D_{\gamma_0}^{(\ell+1)}(\phi - zh_n \kappa) z^{\ell+1} K(z) dz.
\end{aligned}$$

We note that $x \mapsto D_{\gamma_0}^{(\ell+1)}(x)$ is a continuous function for $|x| \leq r < \frac{1}{|\gamma_0|}$, by which we can bound it appropriately in that range. Then, we see that

$$\frac{\int_{-\frac{c}{h_n}}^{\frac{c}{h_n}} [D'_{\gamma_0}(\phi - zh_n) - D'_{\gamma_0}(\phi)] K(z) dz}{h_n^{\ell+1}} \xrightarrow{n \rightarrow \infty} D_{\gamma_0}^{(\ell+1)}(\phi) \int_{\mathbb{R}} z^{\ell+1} K(z) dz < \infty.$$

Hence, the difference is $O(h_n^{\ell+1})$. We remain interested in the convergence of the J_2 term. First, we note that Condition 4 applies to the derivative of K' , as

$$-C_k \leq K(t) \exp(c_K t^2) \leq C_k, \quad \forall |t| \geq t_0.$$

Therefore, we have that $-C_K \leq \lim_{t \rightarrow \infty} K(t) \exp(c_K t^2) \leq C_K$, where we can apply the L'Hôpital's rule to write $-C_K \leq \lim_{t \rightarrow \infty} \frac{K'(t)}{-2c_K t \exp(-c_K t^2)} \leq C_K$. Therefore, for all $\varepsilon > 0$ we can state that there exists t_0^* such that

$$-(1 - \varepsilon) C_k 2c_K t \exp(-c_K t^2) \leq K'(t) \leq C_k 2c_K t \exp(-c_K t^2) (1 + \varepsilon) \quad \forall |t| \geq t_0.$$

Therefore, we can write for J_2 that

$$\begin{aligned}
J_2 &= \frac{1}{n} \sum_{i=1}^n \frac{n}{k} \int_{-\infty}^{\infty} \mathbb{1}\{|zh_n| > c\} \mathbb{1}\left\{Z_i > (\phi - zh_n)a_0\left(\frac{n}{k}\right) + U\left(\frac{n}{k}\right)\right\} \frac{1}{h_n} K'(z) dz \\
&\leq (1 + \varepsilon) C_k \frac{n}{k} \frac{1}{h_n} \int_{\{|zh_n| > c\}} 2c_K \operatorname{sign}(z) \exp(-c_K z^2) dz \\
&= (1 + \varepsilon) 2C_K \frac{n}{k} \frac{1}{h_n} \exp\left(-c_K \left(\frac{c}{h_n}\right)^2\right) \\
&= (1 + \varepsilon) 2C_K \exp\left(-\left[c_K \left(\frac{c}{h_n}\right)^2 - \log(n/k)\right]\right).
\end{aligned}$$

The lower bound can be achieved similarly with a switched sign. Therefore, we can control the term through the assumption on h_n . $\square$

B Generative AI statement

Generative AI was used in this thesis for the following purposes:

1. The code to produce the simulation results was debugged using generative AI.
2. The code to generate the tables and figures was created using generative AI.
3. The optimization procedure was debugged and improved using generative AI.

Generative AI was not used to write text, provide a literature overview, write proofs, come up with new ideas, or a similar intellectual endeavor.